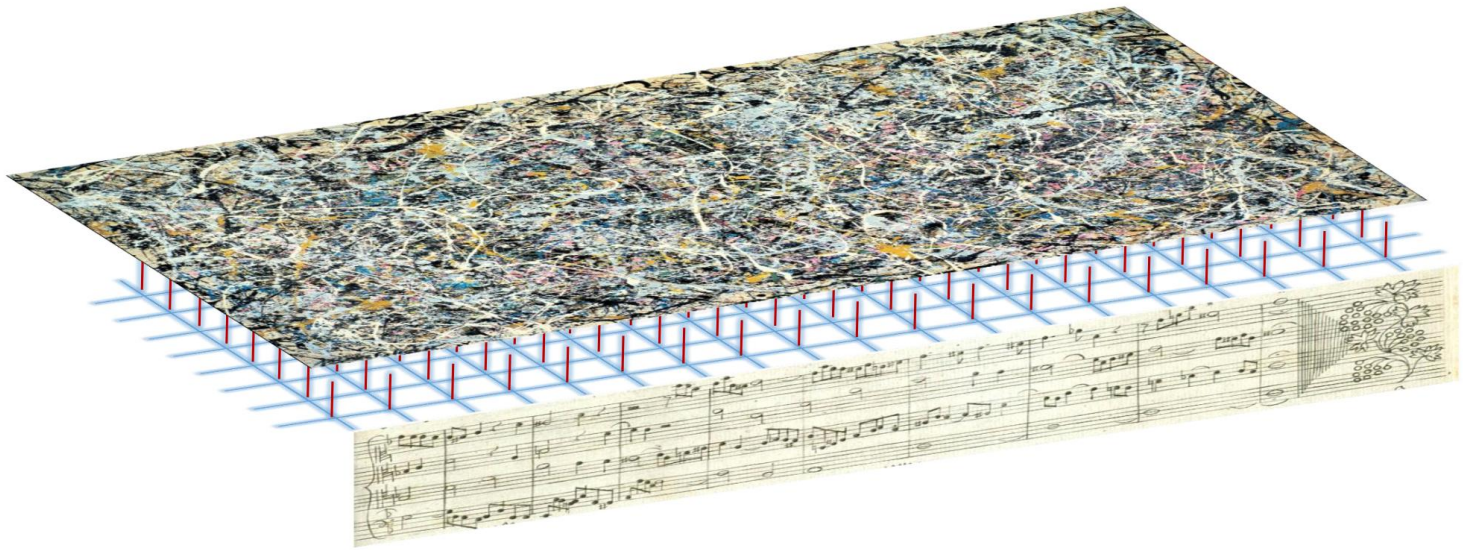
Entanglement as an organizing principle of quantum matter



Frank Verstraete
University of Cambridge & Ghent University
Physics Colloquium, Zagreb, 06/11/2026

The Schrödinger equation



Solvay conference, Brussels 1927

- 1926: Erwin Schrödinger writes down his equation: $i\hbar \frac{\partial}{\partial t} |\Psi\rangle = \hat{H} |\Psi\rangle$
- Summer 1926: Werner Heisenberg solves the helium atom with it
- 1929: Paul Dirac: *"The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble."*

The exponential wall

- Quantum chemistry, statistical physics, condensed matter, nuclear and particle physics: **all the same central problem**
- N particles: the wavefunction lives in a tensor product space

$$|\psi\rangle = \sum_{i_1 i_2 \cdots i_N} \psi_{i_1 i_2 \cdots i_N} \; |i_1\rangle \, |i_2\rangle \cdots |i_N\rangle \;\; \in \;\; \left(\mathbb{C}^d\right)^{\otimes N}$$

- d^N parameters: the exponentiall wall

"In general the many-electron wave function $\Psi(r_1, ..., r_N)$ for a system of N electrons is not a legitimate scientific concept, when $N \geq N_0$, where $N_0 \approx 10^3$."
Walter Kohn, Nobel lecture 1998

- So why is physics possible at all?

How big is the exponential wall?

N spins need 2^N complex amplitudes: every extra spin doubles the memory

10 spins

10^3

amplitudes

16 kilobytes

a 1980s home computer

30 spins

10^9

amplitudes

17 gigabytes

a good laptop

50 spins

10^{15}

amplitudes

18 petabytes

more than any supercomputer

300 spins

10^{90}

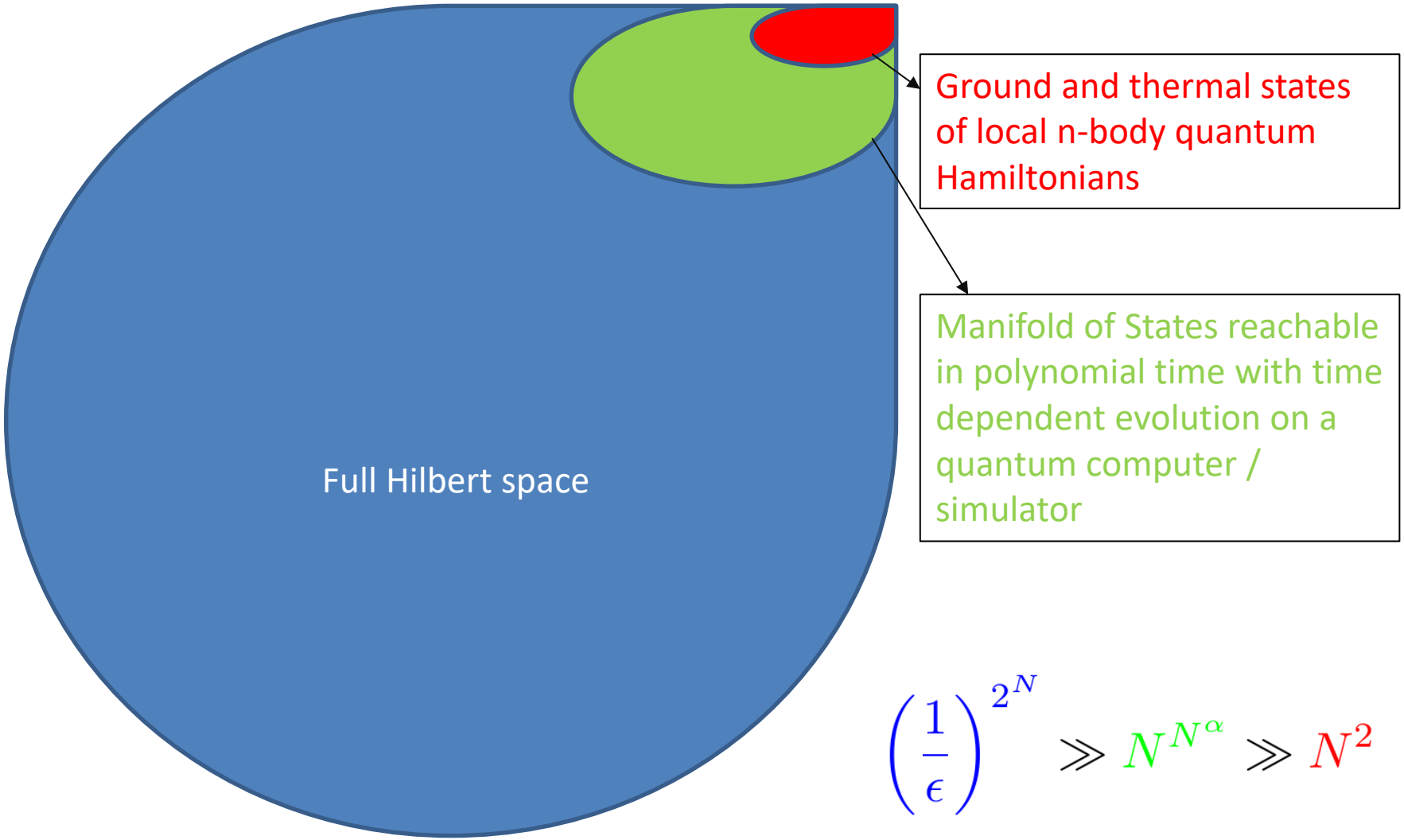
amplitudes

impossible

more amplitudes than atoms in the universe ($\sim 10^{80}$)

Brute force is hopeless beyond ~50 spins. We need to understand where in this enormous space the physical states actually live.

Quantum Simulation of Time-Dependent Hamiltonians and the Convenient Illusion of Hilbert Space



Nuclear Physics B (Proc. Suppl.) 17 (1990) 82–92
North-Holland

AB INITIO QUANTUM CHEMISTRY: A SOURCE OF IDEAS FOR LATTICE GAUGE THEORISTS

Kenneth G. WILSON

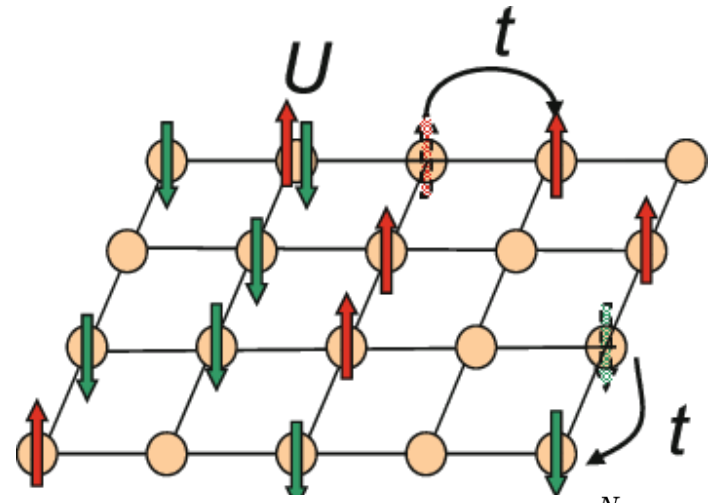
The Ohio State University, Department of Physics, 174 W. 18th Avenue, Columbus, OH 43210 USA

Ab initio quantum chemistry is an emerging computational area that is fifty years ahead of lattice gauge theory, a principal competitor for supercomputer time, and a rich source of new ideas and new approaches to the computation of many fermion systems. An overview of the history, current prospects and future frontiers of quantum chemistry is given, with special emphasis on lessons for lattice gauge theory. Particular reference is given to the role of Gaussian basis functions (in place of grids) and analytic (as opposed to Monte Carlo) methods. The main recommendation to lattice gauge theorists is for greater emphasis on infinite momentum frame studies, using Gaussian basis functions.

In words: Ken Wilson (1990): quantum chemists have been computing forever with many-fermion systems; why is that?

Ways around the wall

1. Hartree-Fock
2. Perturbation theory, Feynman diagrams
3. Density functional theory
4. Quantum Monte Carlo methods
5. Coupled cluster methods
6. Renormalization group, effective field theory

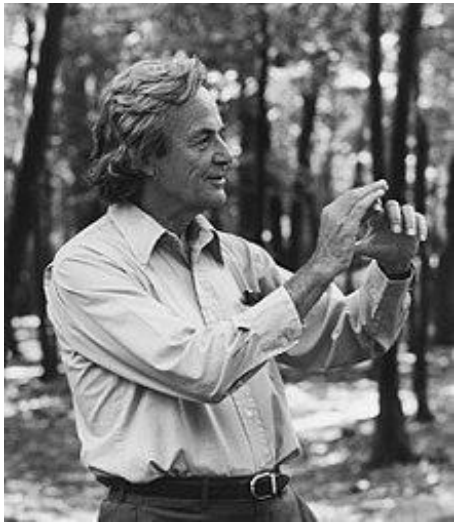


$$\hat{H} = -t \sum_{\langle i,j \rangle, \sigma} (\hat{c}^{\dagger}_{i,\sigma} \hat{c}_{j,\sigma} + \hat{c}^{\dagger}_{j,\sigma} \hat{c}_{i,\sigma}) + U \sum_{i=1}^{N} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$

Hubbard model, 1963

- Amazing progress, but nobody really understands when these methods work
  - the 2D Hubbard model: 60 years old and still not solved

First general solution: quantum computation



"Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy." Feynman 1982

- Use a quantum computer to simulate the Schrödinger equation (Feynman '82, Lloyd '96)
- Turn a bug into a feature:
  - hardness of simulating quantum systems => the power of a quantum computer
  - no-cloning theorem => quantum cryptography

Simulating the Schrödinger equation on a quantum computer

1. Feynman ‘82: a quantum computer is a universal quantum simulator

2. Lloyd ‘96: Trotter decomposition of the time evolution

$$e^{-i\hat{H}t} = \left(e^{-i\hat{H}_1 t/n}\, e^{-i\hat{H}_2 t/n} \cdots e^{-i\hat{H}_m t/n}\right)^n + \mathcal{O}\left(t^2/n\right)$$

   - any local Hamiltonian can be simulated with polynomial resources
   - Lots of progress in optimizing this (Childs, Berry, Low, Chuang, ...)
   - Fermions: Jordan-Wigner, 2D Jordan Wigner, …: duality transformations

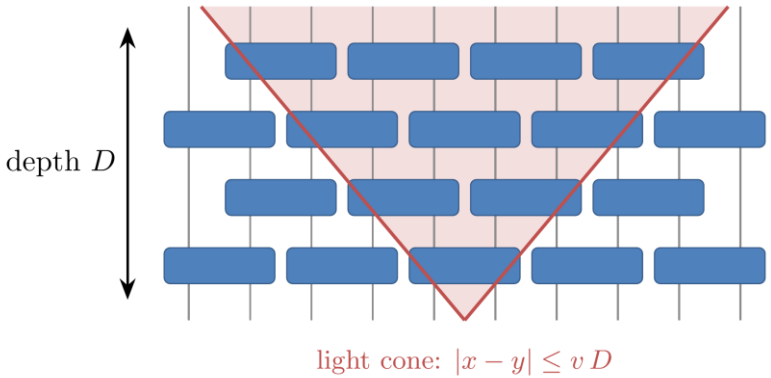
3. Ground states and thermal states:
   - phase estimation (Kitaev), adiabatic evolution (Aharonov et al), dissipative state preparation / quantum Gibbs sampling, Fermionic FFT, …

In words: a quantum computer can imitate any quantum system with local interactions, one small time step after another, at polynomial cost

Kitaev ‘95; Abrams & Lloyd ‘99; Farhi et al. ‘00; Verstraete, Wolf, Cirac ‘09; Temme et al. ‘11; Chen, Kastoryano, Brandão, Gilyén ‘23

Where does the complexity come from?

- **Entanglement**
  - strongly correlated = strongly entangled
  - mean field and perturbation theory miss exactly this regime



- Long-range entanglement gives lower bounds to the depth of any circuit preparing the state

PRL 97, 050401 (2006) PHYSICAL REVIEW LETTERS week ending 4 AUGUST 2006

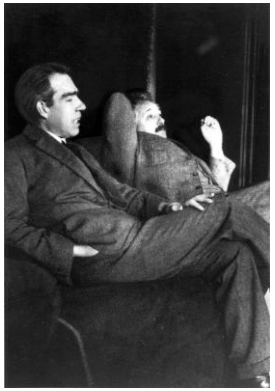
Lieb-Robinson Bounds and the Generation of Correlations and Topological Quantum Order

S. Bravyi,[1] M. B. Hastings,[2] and F. Verstraete[3]

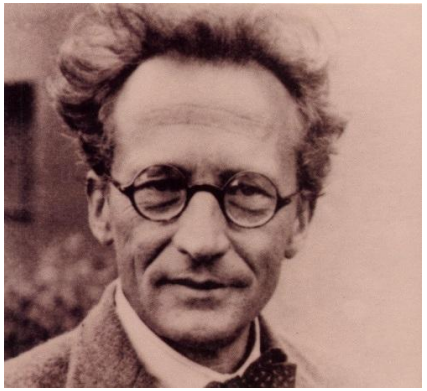
  - topological order => depth grows linearly with the system size

- *Can we turn this around, and use entanglement to describe quantum matter?*

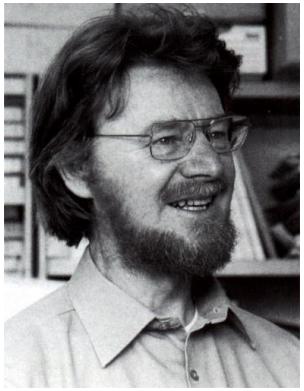
What is entanglement?



Bohr & Einstein
1925



Schrödinger



Bell

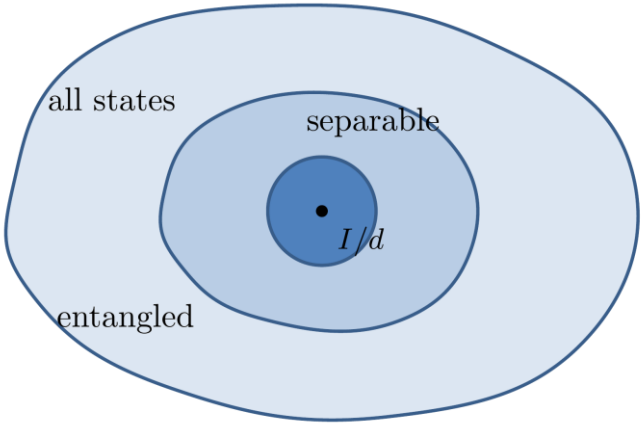
"I would not call that one but rather the characteristic trait of quantum mechanics, the one that enforces its entire departure from classical lines of thought." Schrödinger 1935

- 1935: Einstein, Podolsky, Rosen: is quantum mechanics complete?
- 1935: Schrödinger's cat, and Schrödinger coins the word **entanglement**
- 1964: Bell inequalities, the ultimate test (Nobel prize 2022)
- 1990s: **the** resource that makes a quantum computer more powerful than a classical one
  - **little entanglement => classically simulatable**

Simulating quantum processes with small entanglement

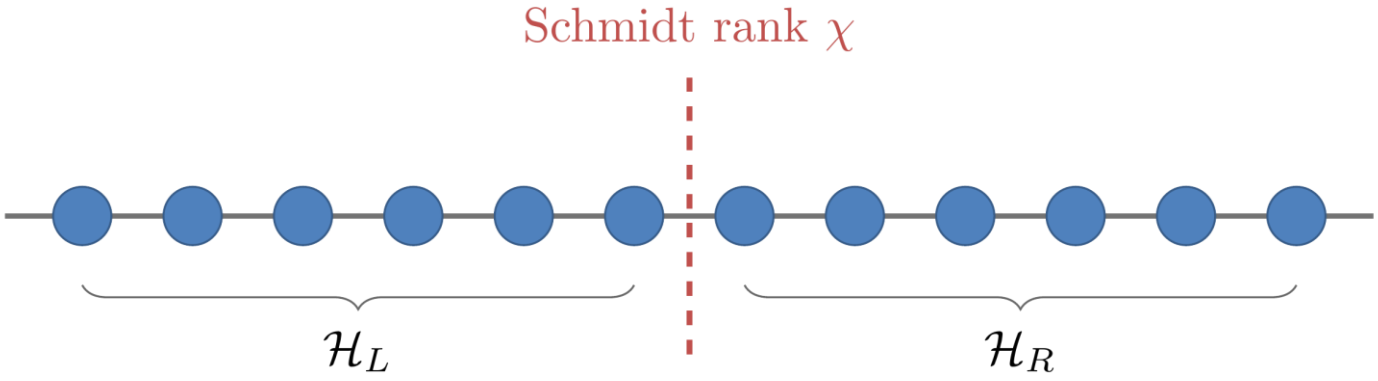
- Aharonov & Ben-Or ‘96: a noisy quantum computer whose state stays separable can be simulated classically

$$\rho_{\text{sep}} \xrightarrow{U,\mathcal{E}} \rho_{\text{sep}} \xrightarrow{U,\mathcal{E}} \rho_{\text{sep}} \xrightarrow{U,\mathcal{E}} \cdots$$



- Simulating slightly entangled quantum computation with tensor networks:





Simulating noisy intermediate quantum computers with tensor networks

arXiv > quant-ph > arXiv:2111.03011

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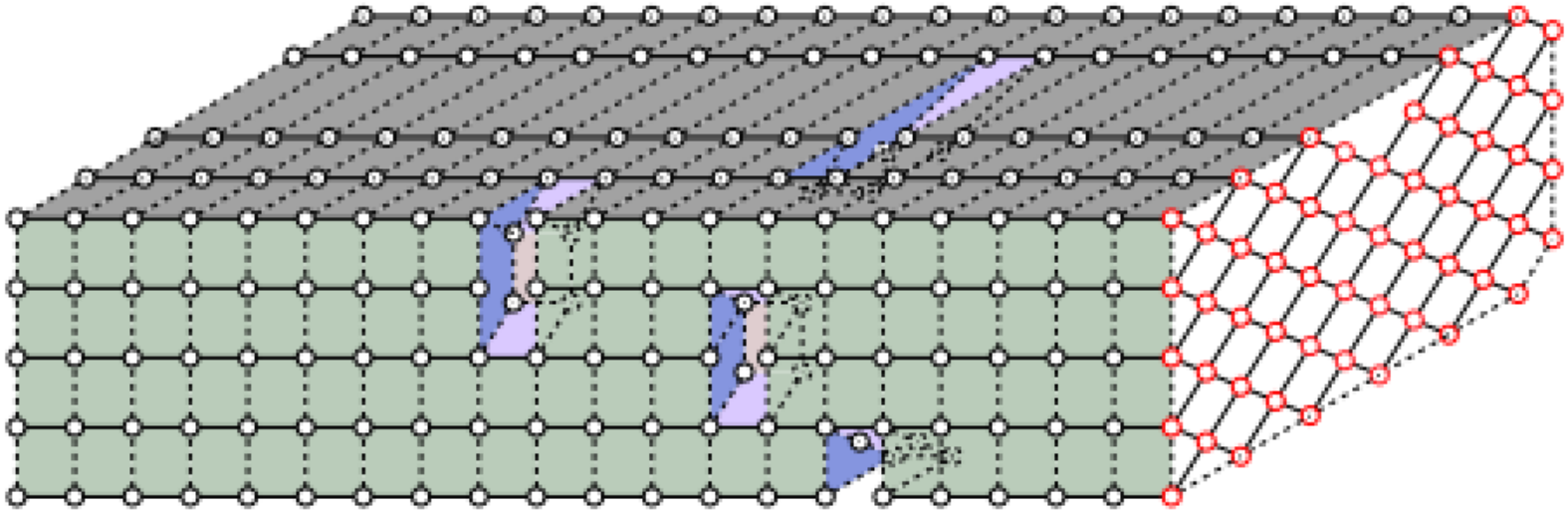
Quantum Physics

[Submitted on 4 Nov 2021]

Solving the sampling problem of the Sycamore quantum supremacy circuits

Feng Pan, Keyang Chen, Pan Zhang

We study the problem of generating independent samples from the output distribution of Google's Sycamore quantum circuits with a target fidelity, which is believed to be beyond the reach of classical supercomputers and has been used to demonstrate quantum supremacy. We propose a new method to classically solve this problem by contracting the corresponding tensor network just once, and is massively more efficient than existing methods in obtaining a large number of uncorrelated samples with a target fidelity. For the Sycamore quantum supremacy circuit with 53 qubits and 20 cycles, we have generated one million uncorrelated bitstrings $\{\mathbf{s}\}$ which are sampled from a distribution $\widehat{P}(\mathbf{s}) = |\widehat{\psi}(\mathbf{s})|^2$, where the approximate state $\widehat{\psi}$ has fidelity $F \approx 0.0037$. The whole computation has cost about 15 hours on a computational cluster with 512 GPUs. The obtained one million samples, the contraction code and contraction order are made public. If our algorithm could be implemented with high efficiency on a modern supercomputer with ExaFLOPS performance, we estimate that ideally, the simulation would cost a few dozens of seconds, which is faster than Google's quantum hardware.

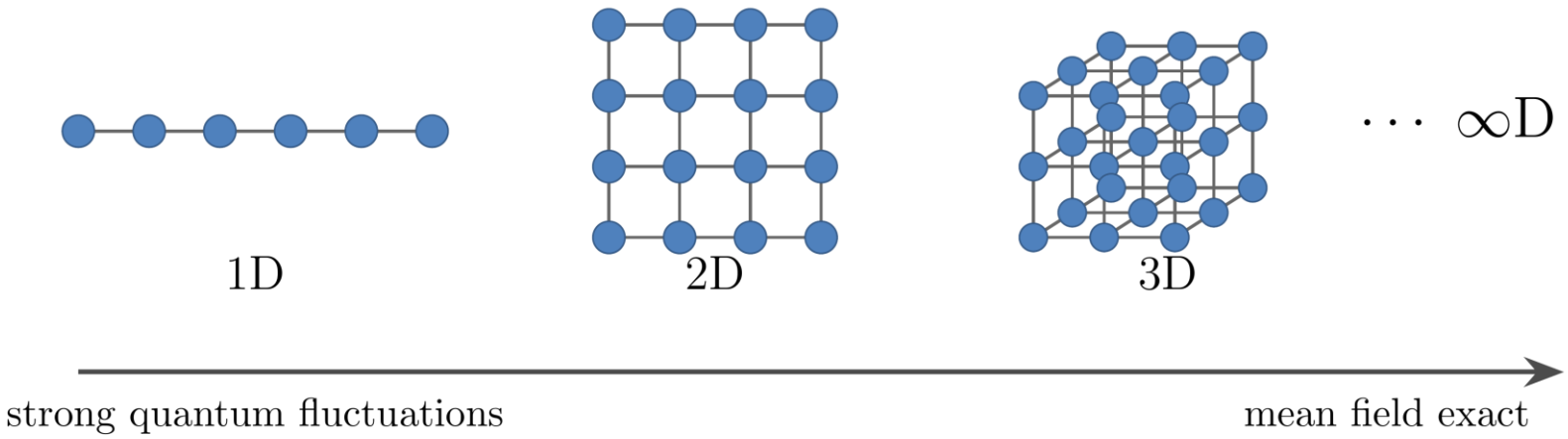


Does that teach us how to simulate many-body systems?

- **Yes.** High dimensions: mean field theory

$$|\psi\rangle \approx |\phi\rangle \otimes |\phi\rangle \otimes \cdots \otimes |\phi\rangle$$

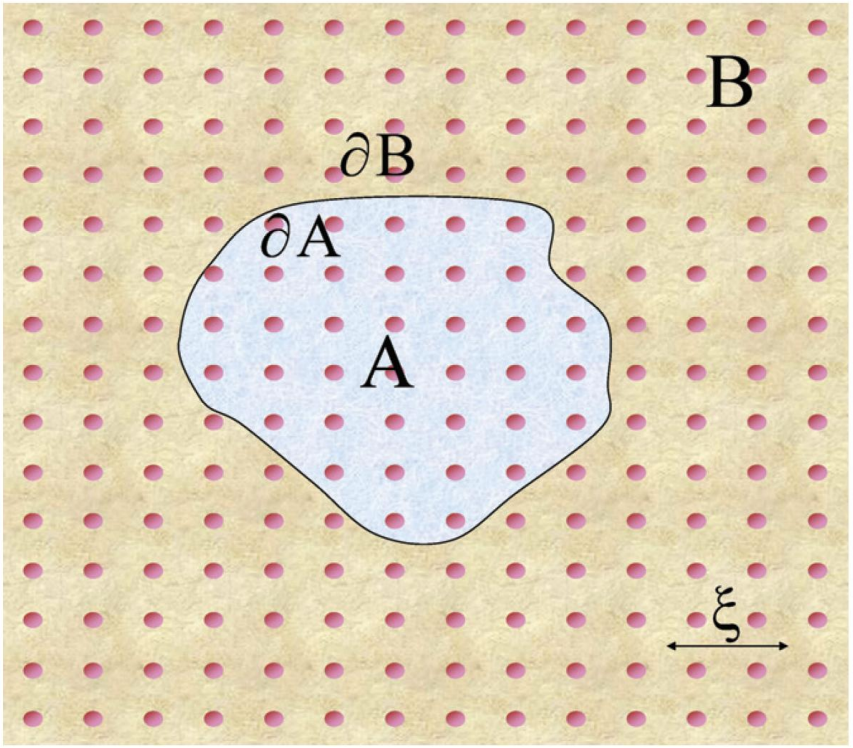
- **Maybe**. Lower dimensions



- The interesting physics happens in 1D and 2D: Luttinger liquids, fractional quantum Hall effect, anyons, high-T_c superconductivity, spin liquids, ...
  - **it is all about entanglement**

Area Laws for the entanglement entropy

- Ground and Gibbs states of interacting quantum many body Hamiltonians with ***local interactions*** have very peculiar properties
  - Area law for the entanglement entropy (ground states) or for mutual information (Gibbs states)
  - "explains" why physics is possible at all



Ground states:

$$S(\rho_A) = c.\partial A$$

Srednicki '93; Hastings '07; …

$$S(\rho_A) = \frac{c}{6}.\log(A/\epsilon)$$

Holzhey, Larsen, Wilczek '94; …

Gibbs states:

$$\begin{aligned} I(A,B) &= S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \\ &= c.\partial A \end{aligned}$$

Wolf, Hastings, FV, Cirac '08

Entanglement in an antiferromagnet

- Heisenberg antiferromagnet (spin ½):

$$\mathcal{H} = \sum_i \vec{S}_i \vec{S}_{i+1}$$

- What is the ground state?
  - 2 spins:  $(|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle)$
  - 3 spins:



$(|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle)|\uparrow\rangle$ –

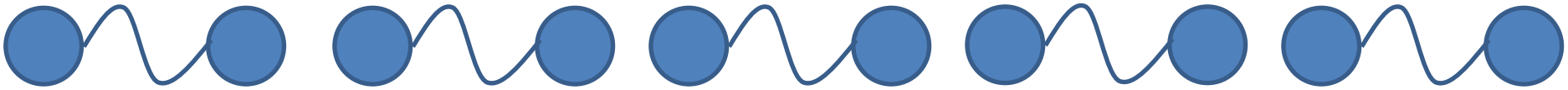


$|\uparrow\rangle(|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle)$

- Monogamy of entanglement: impossibility of sharing a singlet with two spin ½'s. Emerging phenomena arise because of this competition.

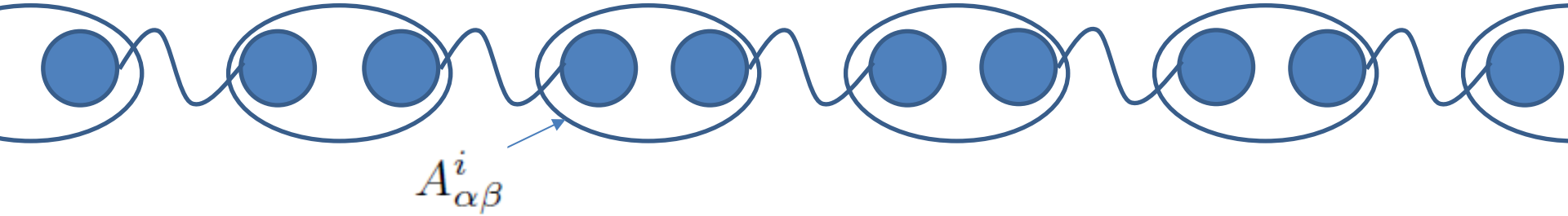
Tensors for Quantum Spin Chains

- Build the state from entangled pairs between neighbours, then let a small tensor A map the two virtual spins on each site onto one physical spin: ***matrix product states*** ***(MPS)***



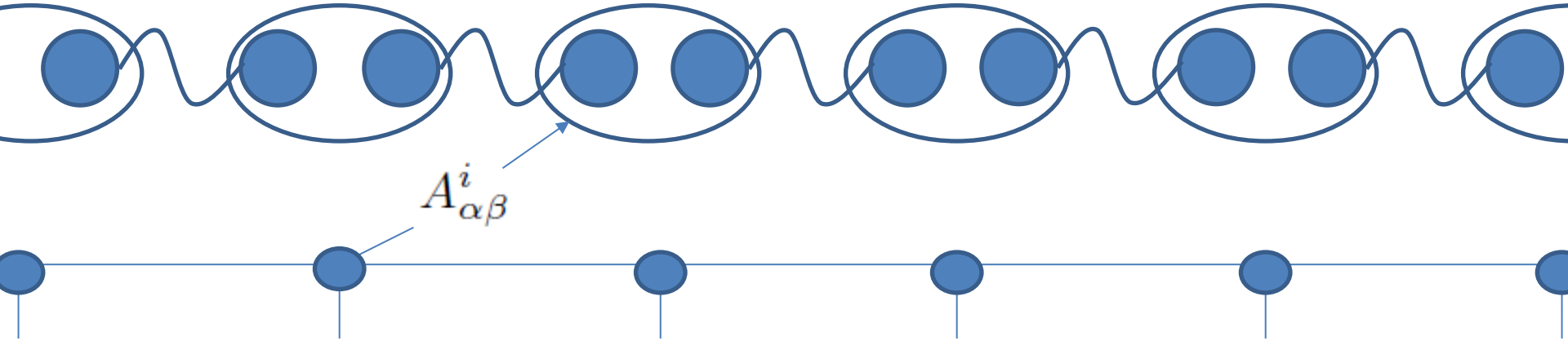
Tensors for Quantum Spin Chains

- Build the state from entangled pairs between neighbours, then let a small tensor A map the two virtual spins on each site onto one physical spin: ***matrix product states (MPS)***



Tensors for Quantum Spin Chains

- Build the state from entangled pairs between neighbours, then let a small tensor A map the two virtual spins on each site onto one physical spin: ***matrix product states (MPS)***



$$|\psi\rangle = \sum_{i_1 i_2 i_3 \cdots} \mathrm{Tr}\left(A^{i_1} A^{i_2} A^{i_3} \cdots\right) |i_1\rangle |i_2\rangle |i_3\rangle \cdots$$

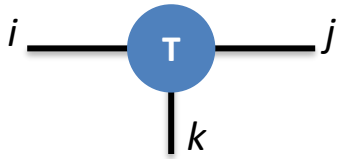
Tensor networks: a picture language



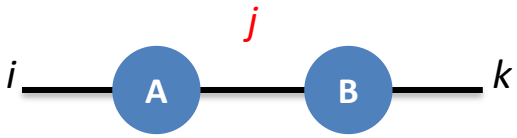
vector v_i



matrix M_{ij}



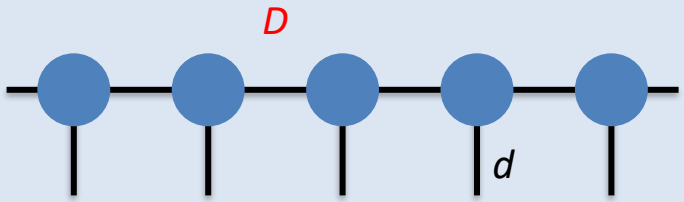
tensor T_{ijk}



$= \Sigma_j \, A_{ij} \, B_{jk} = (AB)_{ik}$

Connected legs are summed over: matrix multiplication, drawn as a picture

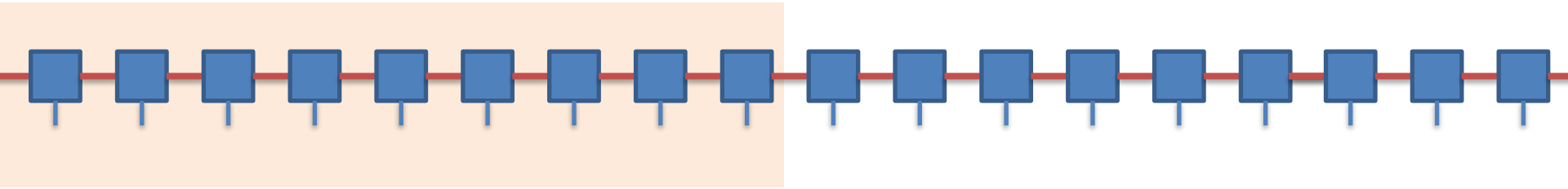
Matrix product state

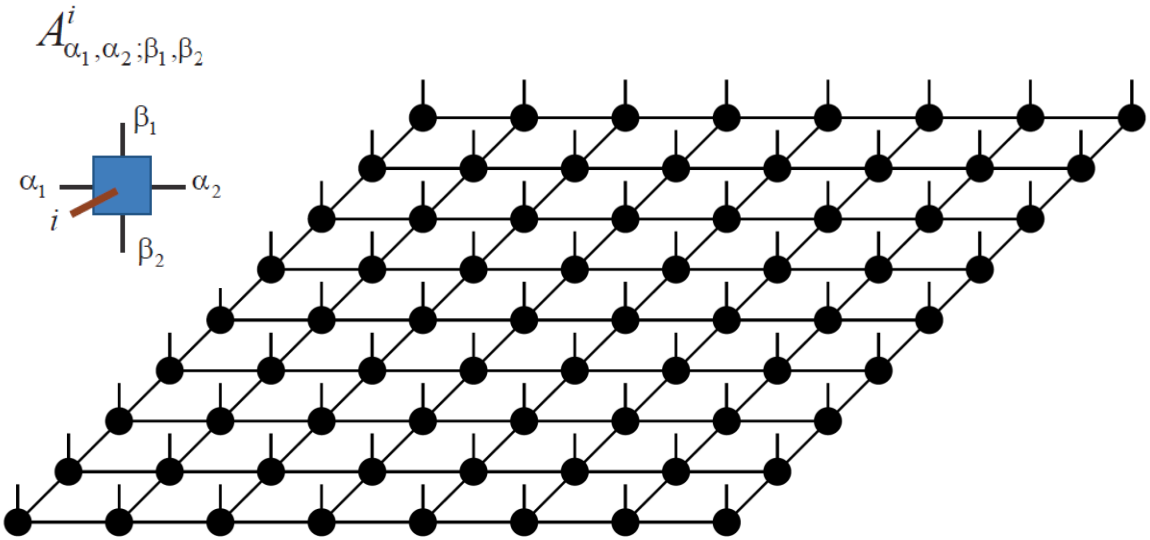


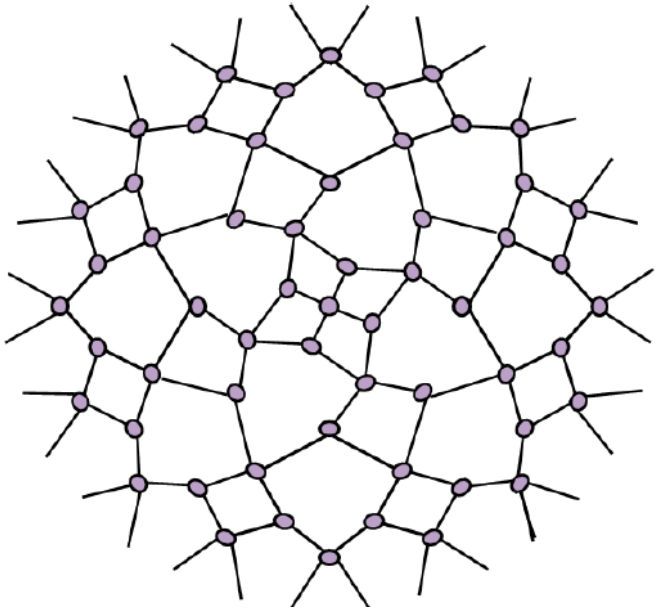
one small tensor per site

- Full wavefunction: d^N numbers
- MPS: $N \cdot d \cdot D^2$ numbers
- 300 spins, D = 100: **6 million numbers instead of 10^{90}**
- Entanglement across any cut ≤ log D: the area law is built in

Entanglement as building block of matter: Quantum Tensor Networks















- MPS form a smooth, low-dimensional surface in the huge Hilbert space: a Kähler manifold (complex + symplectic)

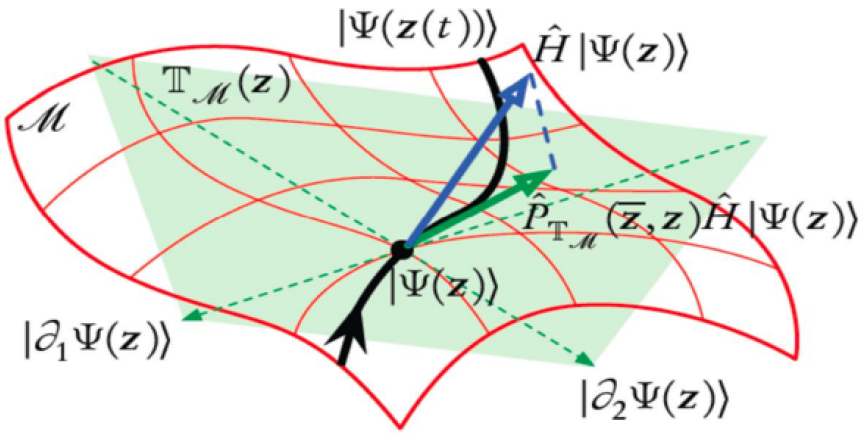
Geometry of matrix product states: Metric, parallel transport, and curvature, Haegeman, … , FV ’14

  - Ground states of gapped quantum spin chains correspond one-to-one to points on this surface

FV, Cirac ‘06; Hastings ‘07

  - The symplectic structure gives Poisson brackets: quantum time evolution on this surface becomes a small set of *classical* equations of motion (TDVP)

Time-Dependent Variational Principle for MPS, Haegeman,…,FV ’12; ‘16



$$(i)\frac{d}{dt}|\psi\rangle = -\mathcal{P}_{\mathcal{T}(A)}\hat{H}|\psi\rangle$$

Tensor networks: central concepts

1 Model the entanglement

Tensors encode the **entanglement pattern**: modelling correlations makes much more sense than modelling the wavefunction directly

2 Efficient by construction

Thanks to the area law, networks can be contracted efficiently: a **holographic** map from 3D → 2D → 1D → 0D problems

3 Infinite systems for free

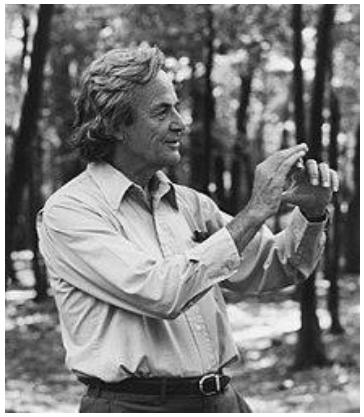
MPS live directly in the thermodynamic limit. For critical systems, finite-size scaling becomes **finite-entanglement scaling**

4 One tensor knows all

Phases of matter, including topological phases and gauge theories, are distinguished by the **symmetries of a single tensor**

Feynman's dream

"Now, in field theory, what's going on over here and what's going on over there and all over space is more or less the same. Why do we have to keep track in our functional of all things going on over there while we are looking at the things that are going on over here? …. It's really quite insane actually: we are trying to find the energy by taking the expectation of an operator which is located here and we present ourselves with a functional which is dependent on everything all over the map. That's something wrong. Maybe there is some way to surround the object, or the region where we want to calculate things, by a surface and describe what things are coming in across the surface. It tells us everything that's going on outside.
I'm talking about a new kind of idea but that's the kind of stuff we shouldn't talk about at a talk, that's the kind of stuff you should actually do!"



"Difficulties in Applying the Variational Principle to Quantum Field Theories", Wangerooge 1987, Proceedings,
Variational calculations in quantum field theory

Overview

1. Why is the quantum many-body problem so hard?

2. Tensor networks: what can we compute?

3. Tensor networks: what do they teach us about quantum matter?

Density Matrix Formulation for Quantum Renormalization Groups

Steven R. White

Department of Physics, University of California, Irvine, California 92717

(Received 22 May 1992)

A generalization of the numerical renormalization-group procedure used first by Wilson for the Kondo problem is presented. It is shown that this formulation is optimal in a certain sense. As a demonstration of the effectiveness of this approach, results from numerical real-space renormalization-group calculations for Heisenberg chains are presented.

TABLE I. Ground-state energies per site of infinite $S=\frac{1}{2}$ and $S=1$ antiferromagnetic Heisenberg chains. The exact Bethe-ansatz result for the energy of the $S=\frac{1}{2}$ chain is $-\ln 2+\frac{1}{4}=-0.443\,147\ldots$, and m is the number of states kept in block A (counting a triplet as three states, etc.). Results labeled ∞ are obtained from a linear extrapolation to $P_m \to 1$. Monte Carlo results are taken from Refs. [7] and [5].

m	$S=\frac{1}{2}$ $E_0-E_0^{\text{exact}}$	$S=\frac{1}{2}$ $1-P_m$	$S=1$ $-E_0$	$S=1$ $1-P_m$
16	5.8×10^{-5}	8.0×10^{-6}	1.401 089	4.8×10^{-5}
24	1.7×10^{-5}	1.9×10^{-6}	1.401 380	1.6×10^{-5}
36	7.8×10^{-6}	9.0×10^{-7}	1.401 437	6.6×10^{-6}
44	3.2×10^{-6}	3.6×10^{-7}	1.401 476	1.1×10^{-6}
∞	1.9×10^{-7}		1.401 484(2)	
MC	$\sigma=5\times10^{-4}$		1.401 5(5)	

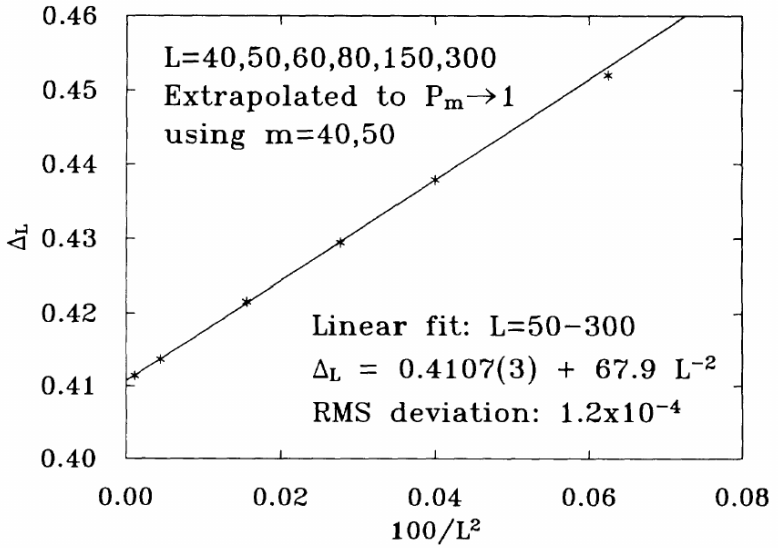
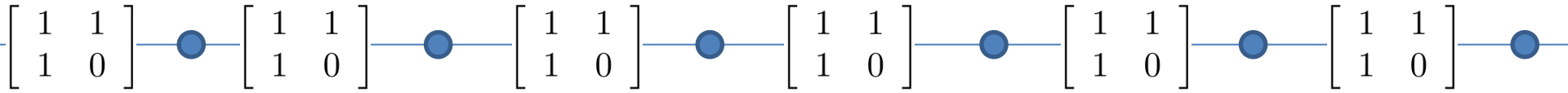


FIG. 2. The Haldane gap as a function of lattice size L.

In words: DMRG (1992), the first tensor network method, solved 1D quantum chains

Counting: hard square constants

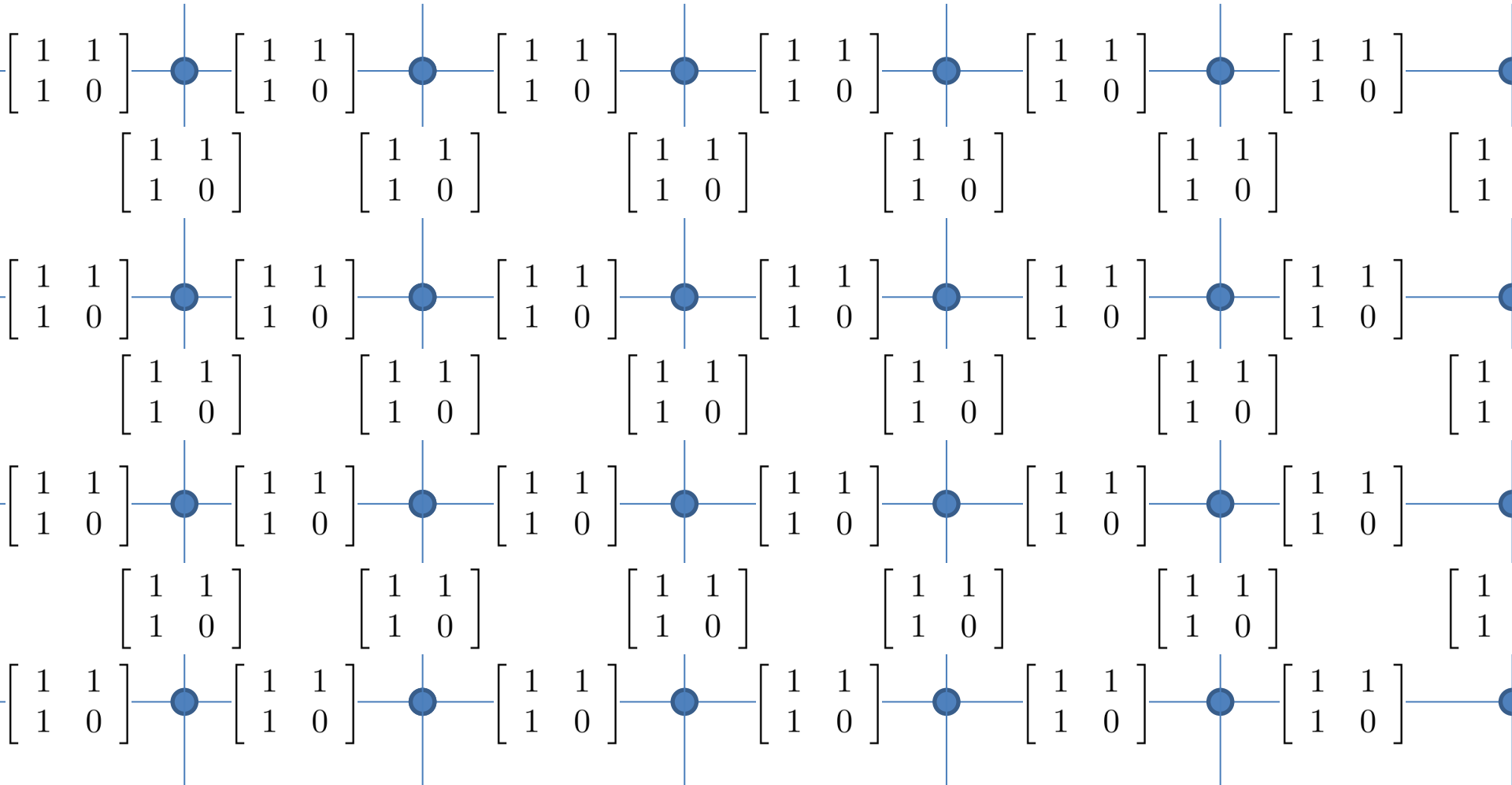
- 1-dimension:
  - count number of configurations of bits such that a 1 is surrounded by 0's
  - Transfer matrix approach: evaluate following tensor network



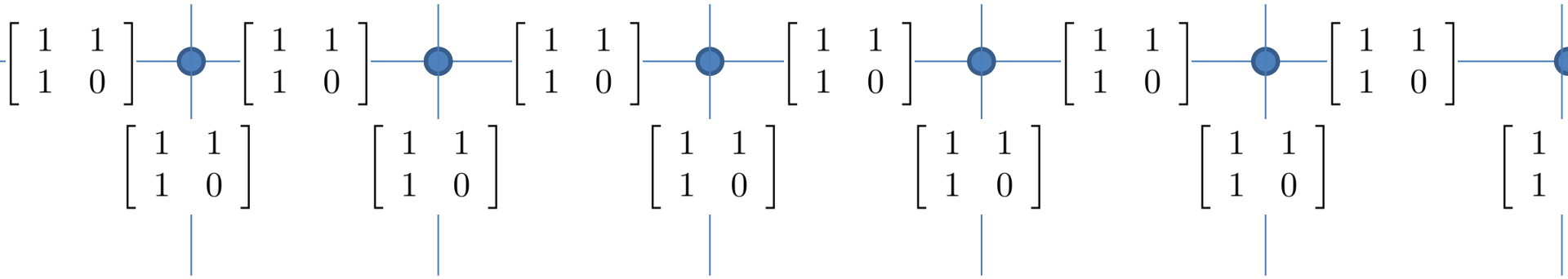
  - Number of configurations is: $\mathrm{Tr}\left(\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^L\right) = \left(\frac{1+\sqrt{5}}{2}\right)^L + \left(\frac{1-\sqrt{5}}{2}\right)^L$

In words: counting becomes multiplying small matrices, a tiny tensor network. The number of configurations grows like the golden ratio, 1.618…, to the power L

- 2 dimensions: contraction of 2D tensor network yields # configurations:



- Problem is reduced to finding leading eigenvalue of transfer matrix / MPO:

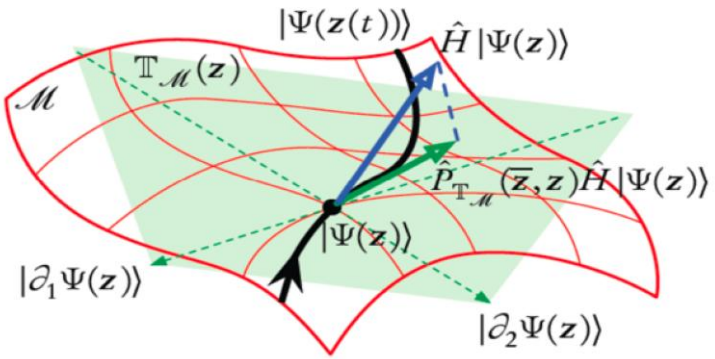


2	1.5030477
4	1.50304808246
6	1.50304808247533218
8	1.5030480824753322642
10	1.503048082475332264322058
20	1.50304808247533226432206632947554
30	1.503048082475332264322066329475553689377
Baxter[128]	1.5030480824753322643220663294755536893857 81
40	1.503048082475332264322066329475553689385781038609
50	1.503048082475332264322066329475553689385781038610303
60	1.503048082475332264322066329475553689385781038610305061
70	1.503048082475332264322066329475553689385781038610305062026556
80	

Table 1: Free energy of the hard squares model; with bond dimension $D = 80$, we get 58 digits of precision.

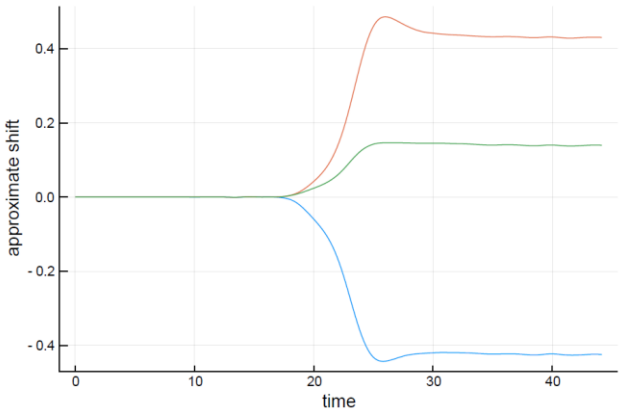
arXiv:1611.08519

MPS: computational aspects

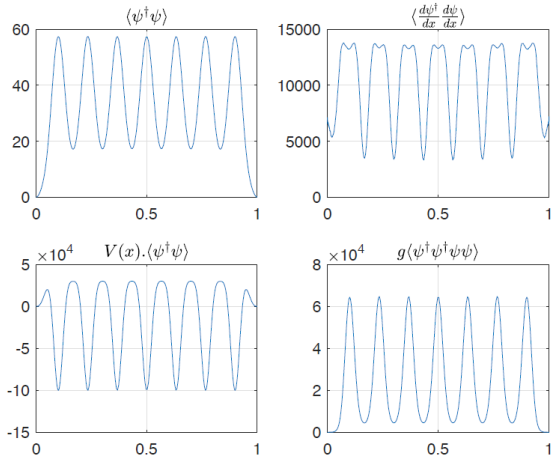


$$(i)\frac{d}{dt}|\psi\rangle = -\mathcal{P}_{\mathcal{T}(A)}\hat{H}|\psi\rangle$$

Time-Dependent Variational Principle for MPS
PRL 107, 70601



Scattering of magnons in spin 1 Heisenberg model
PRR 3, 013078



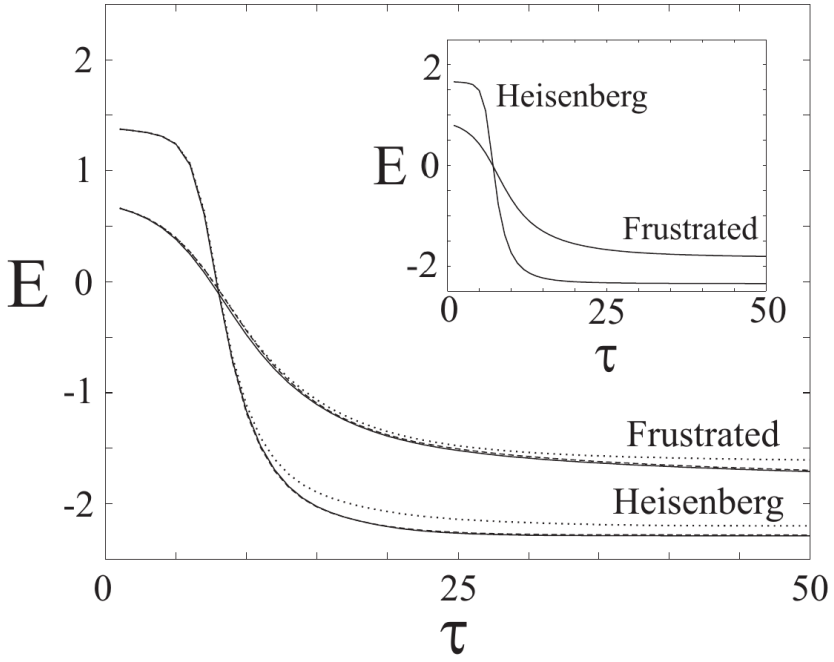
Interacting bosons in the continuum: cMPS and quantum Gross-Pitaevskii
PRL 128, 020501

Renormalization algorithms for Quantum-Many Body Systems in two and higher dimensions

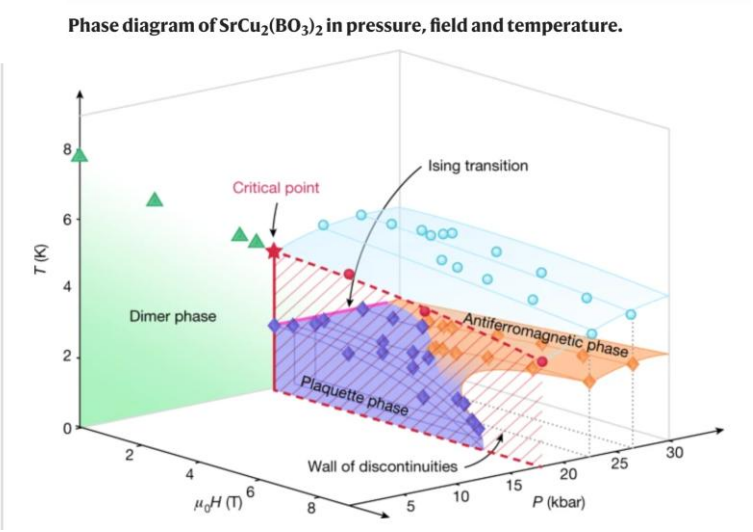
F. Verstraete and J. I. Cirac
Max-Planck-Institut für Quantenoptik, Hans-Kopfermann-Str. 1, Garching, D-85748, Germany.

arXiv:cond-mat/0407066

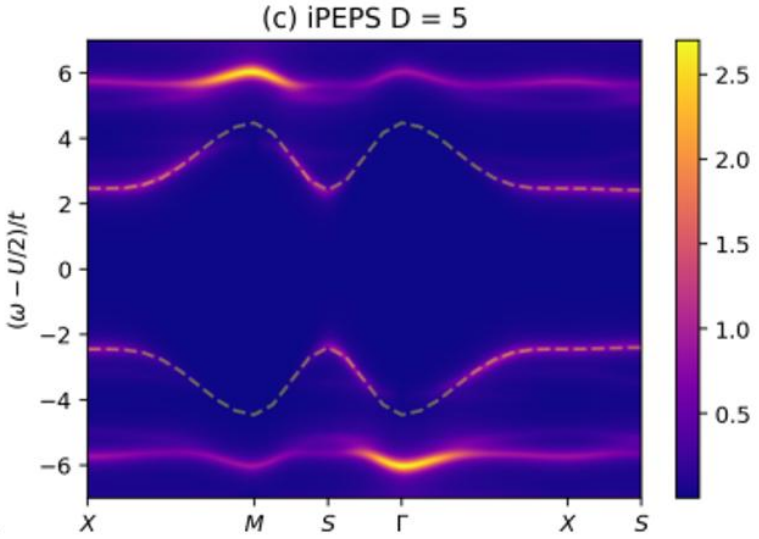
We describe quantum many–body systems in terms of projected entangled–pair states, which naturally extend matrix product states to two and more dimensions. We present an algorithm to determine correlation functions in an efficient way. We use this result to build powerful numerical simulation techniques to describe the ground state, finite temperature, and evolution of spin systems in two and higher dimensions.



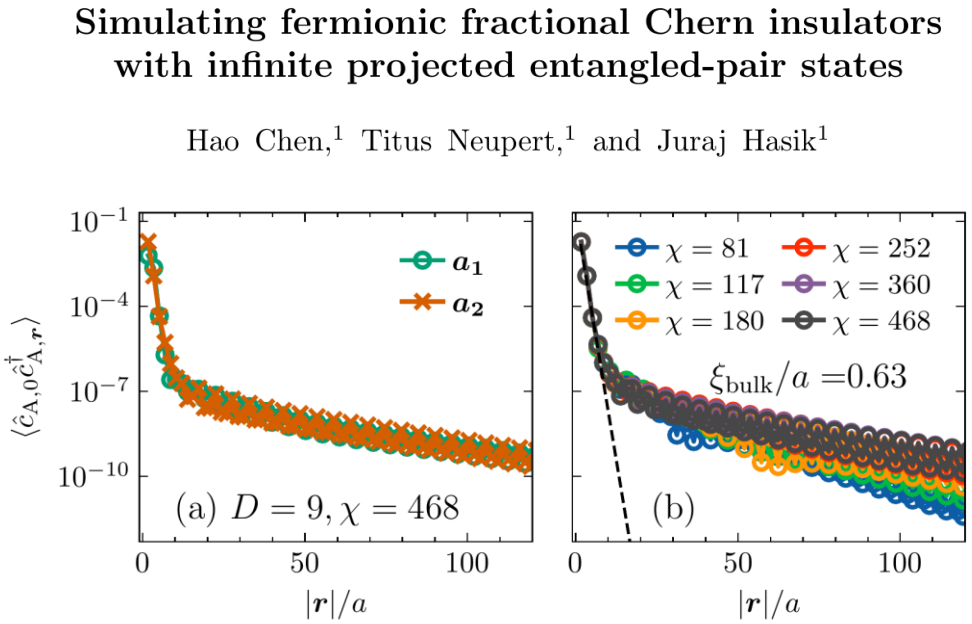
In words: PEPS (2004) extend matrix product states to two and more dimensions



Shastry-Sutherland model
Mila, Corboz et al., Nature 592, 370 (2021)

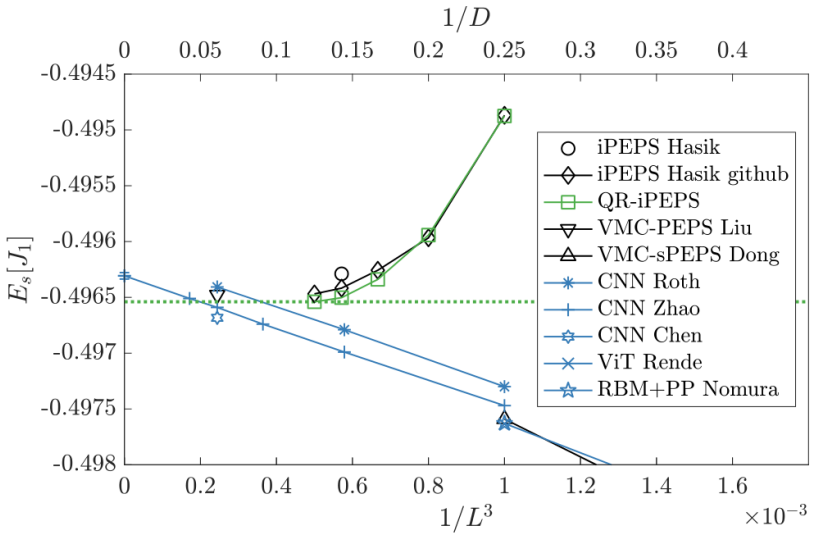


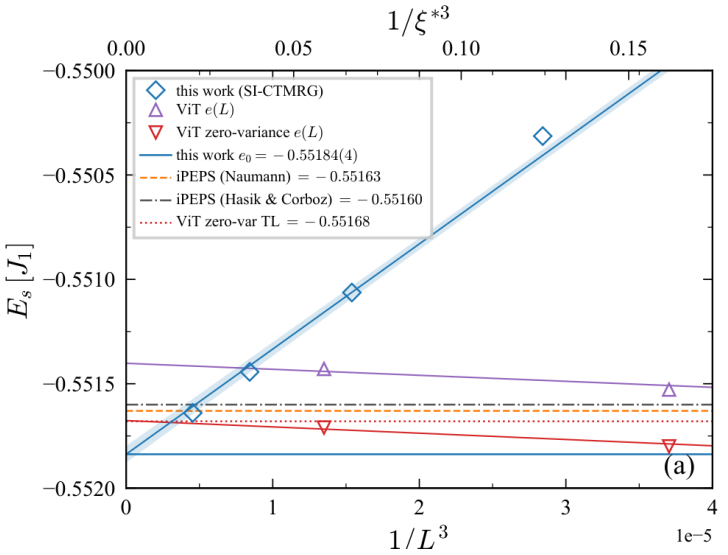
Quasi-particle energies in the 2D Hubbard model:
Ponsioen, Assaad, P. Corboz; SciPost Physics, 2022



Frustrated quantum Heisenberg models

Corboz group (Amsterdam): 100× speed-ups on GPUs





Frustrated J1-J2 model: D = 8 and χ = 700 give the lowest variational energy to date, in 54 minutes on one H100 GPU

Zhang, Yang & Corboz, PRB 113, L201106 (2026)

Triangular-lattice Heisenberg: D = 4 … 8, extrapolated $e_\infty = -0.55184(4)$, about 10 hours on one H100 GPU

Zhang & Corboz, arXiv:2607.15158 (2026)

Hubbard model with PEPS at D = 28

W.-Y. Liu, H. Zhai, R. Peng, Z.-C. Gu & G. K.-L. Chan (Caltech), *Phys. Rev. Lett. 134, 256502 (2025)*

D = 28

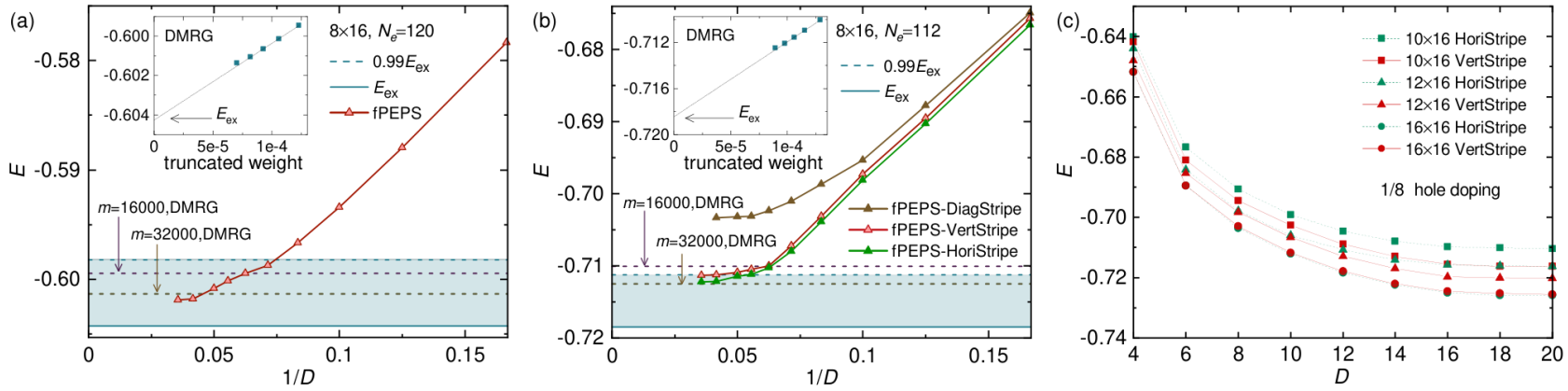
bond dimension

16 × 16

lattices at 1/8 hole doping, beyond the reach of DMRG

< DMRG

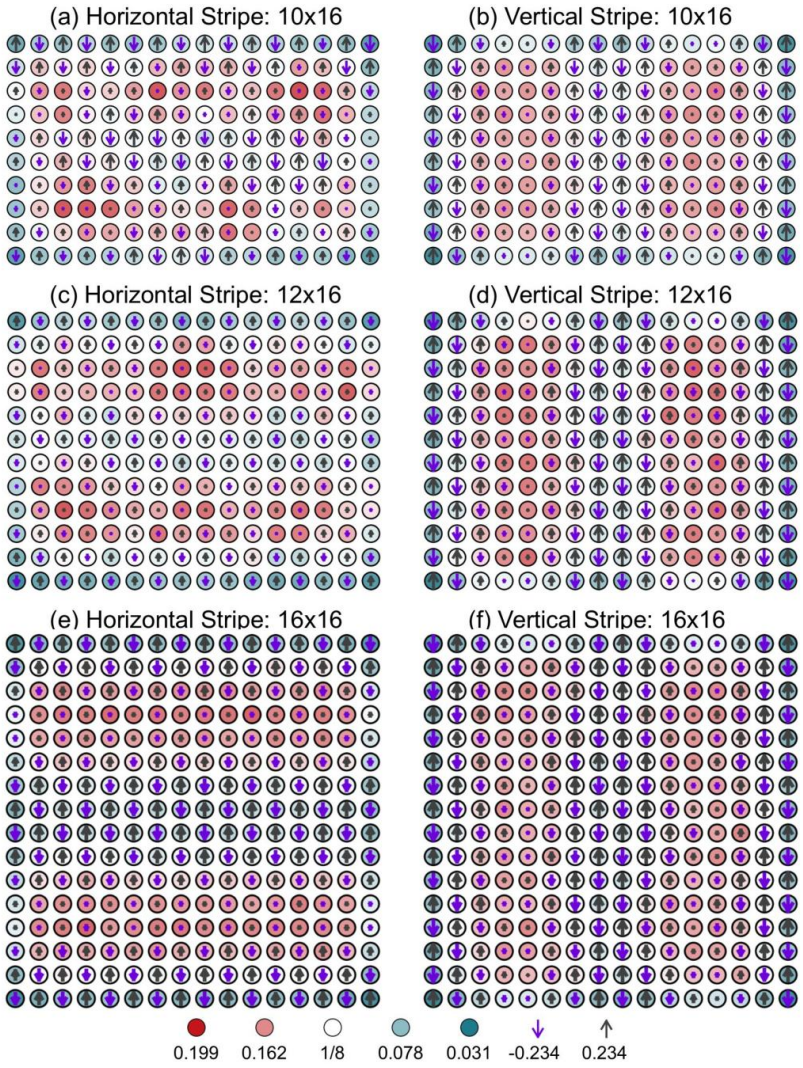
lower energies than DMRG with 32 000 SU(2) multiplets



Energy per site of 8 × 16 Hubbard lattices (U = 8) vs. 1/D, compared with the largest DMRG runs

In words: Monte Carlo sampling makes huge bond dimensions affordable, and PEPS now beat the best simulations on the 2D Hubbard model

Hubbard stripes on 16 × 16 lattices



Entanglement scaling

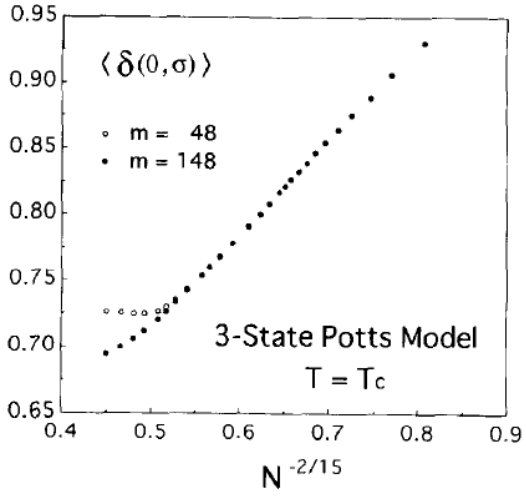


- When simulating a critical theory with finite bond dimension and on an infinite lattice, you effectively introduced a gap that depends on the bond dimension

  => scaling theory

Nishino, Okunishi, Kikuchi '96:

$$M(N) = N^{-(d-2+\eta)/2} g(\xi(m)/N)$$



Pollmann, Mukerjee, Turner, Moore '09; Pirvu, Tagliacozzo, FV, Vidal '12

$$E_0(\xi_\epsilon) = E_0(\infty) + \frac{A}{\xi_\epsilon^2}$$

$$E_0(\xi_D) = E_0(\infty) + \frac{\beta}{\xi_D} P_r(b, D)$$

$$\xi(D) \propto D^\kappa \qquad \kappa = \frac{6}{c\left(1 + \sqrt{\frac{12}{c}}\right)}$$

Scaling hypothesis for MPS

- simultaneous scaling in the distance to the critical point and in the bond dimension:

$$\delta = \sum_{i=1}^{n} c_i \epsilon_i,$$

Rams, Czarnik & Cincio '18

$$\delta^{-\beta/\nu} m(t,\delta) = m(\delta^{-1/\nu} t, 1) = \tilde{m}(\delta^{-1/\nu} t)$$

$$\delta \xi(t,\delta) = \xi(\delta^{-1/\nu} t, 1) = \tilde{\xi}(\delta^{-1/\nu} t)$$

$$\exp\left(\frac{6}{c} S(t,\delta)\right) = s \exp\left(\frac{6}{c} S(s^{-1/\nu} t, s\delta)\right)$$

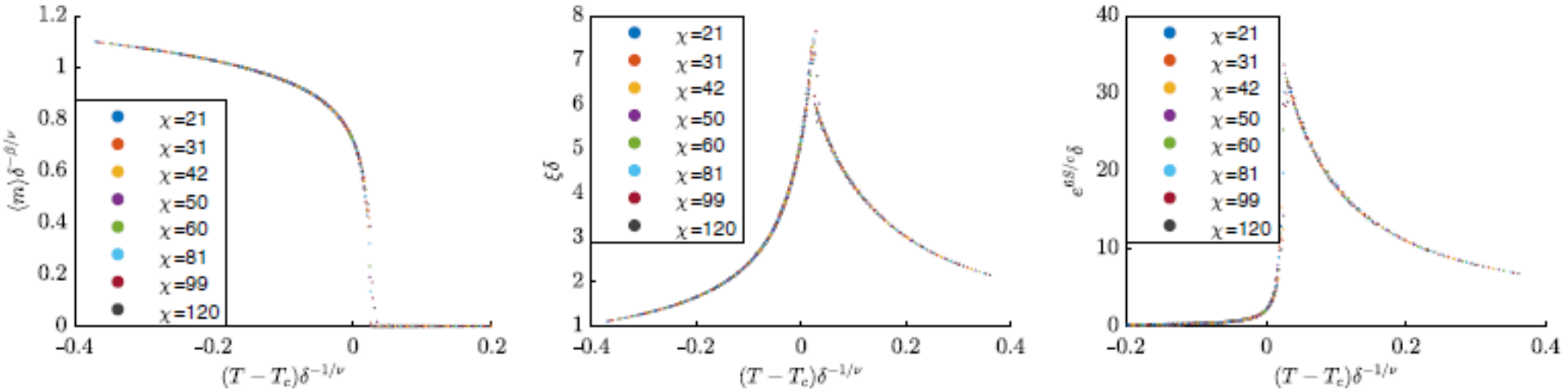
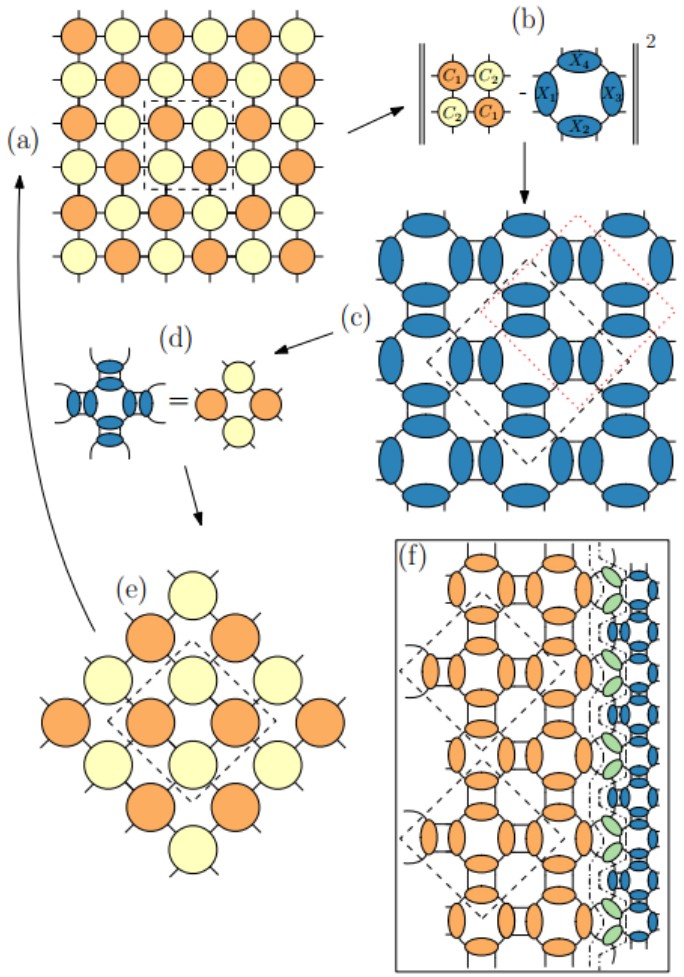


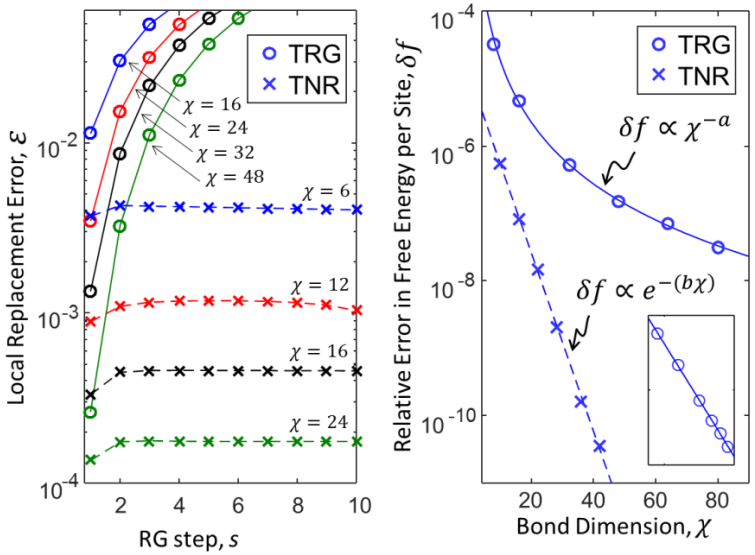
FIG. 2. Collapse plots for the Potts model, calculated with MPS of bond dimension 21,31,42,50,60,81,99, and 120, for 96 different temperatures linearly spaced between $T = 0.9939$ and $T = 0.9954$. Left, magnetization; middle, correlation length; right, bipartite entanglement entropy.

Vanhecke, Haegeman, Van Acoleyen, Vanderstraeten &FV '19

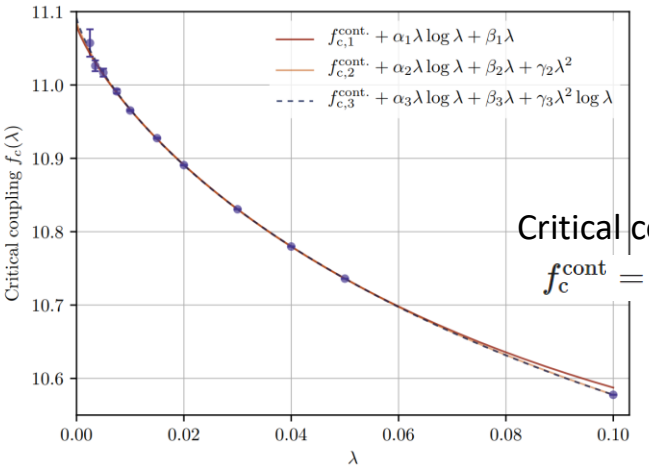
Real-space renormalization with tensor networks: coarse-grain the network step by step (MERA, TRG, TNR, Loop TNR)

Vidal, Evenbly; Gu, Wen; Xiang et al.; Bal et al; Kawashima et al.; Kennedy, Rychkov; …





Tensor Network Renormalization
Evenbly, Vidal PRL 115, 180405



Critical coupling cst:

$f_c^{\text{cont}} = 11.0861\,(90)$

Computing the renormalization group flow of two-dimensional Φ^4 theory with tensor networks
Delcamp, Tilloy PRR 2, 033278

Tensor renormalization group study of cold and dense QCD in the strong coupling limit

Yuto Sugimoto,[1,*] Shinichiro Akiyama,[2,3,†] and Yoshinobu Kuramashi[2,‡]

TABLE I. Values of m, μ_c^n, $\Delta\langle n\rangle(m)$, μ_c^χ and $\Delta\langle\bar{\chi}\chi\rangle(m)$ at the first-order phase transition. All the results are obtained with $(D, D_{3d}) = (55, 120)$.

m	μ_c^n	$\Delta\langle n\rangle$	μ_c^χ	$\Delta\langle\bar{\chi}\chi\rangle$
1.900	1.5315(5)	0.6326(34)	1.532(1)	0.6925(13)
1.925	1.5405(5)	0.589(5)	1.541(1)	0.6388(15)
1.950	1.5505(5)	0.538(7)	1.550(1)	0.5703(21)
1.975	1.5595(5)	0.488(8)	1.559(1)	0.473(5)
2.000	1.5685(5)	0.425(12)	1.569(1)	0.390(6)
2.025	1.5775(5)	0.350(18)	1.578(1)	0.308(6)

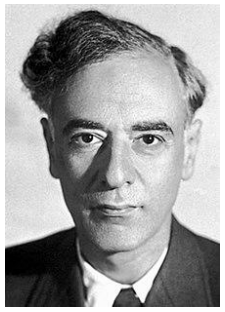
Overview

1. Why is the quantum many-body problem so hard?

2. Tensor networks: what can we compute?

3. Tensor networks: what do they teach us about quantum matter?

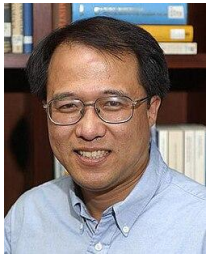
Phases of Matter

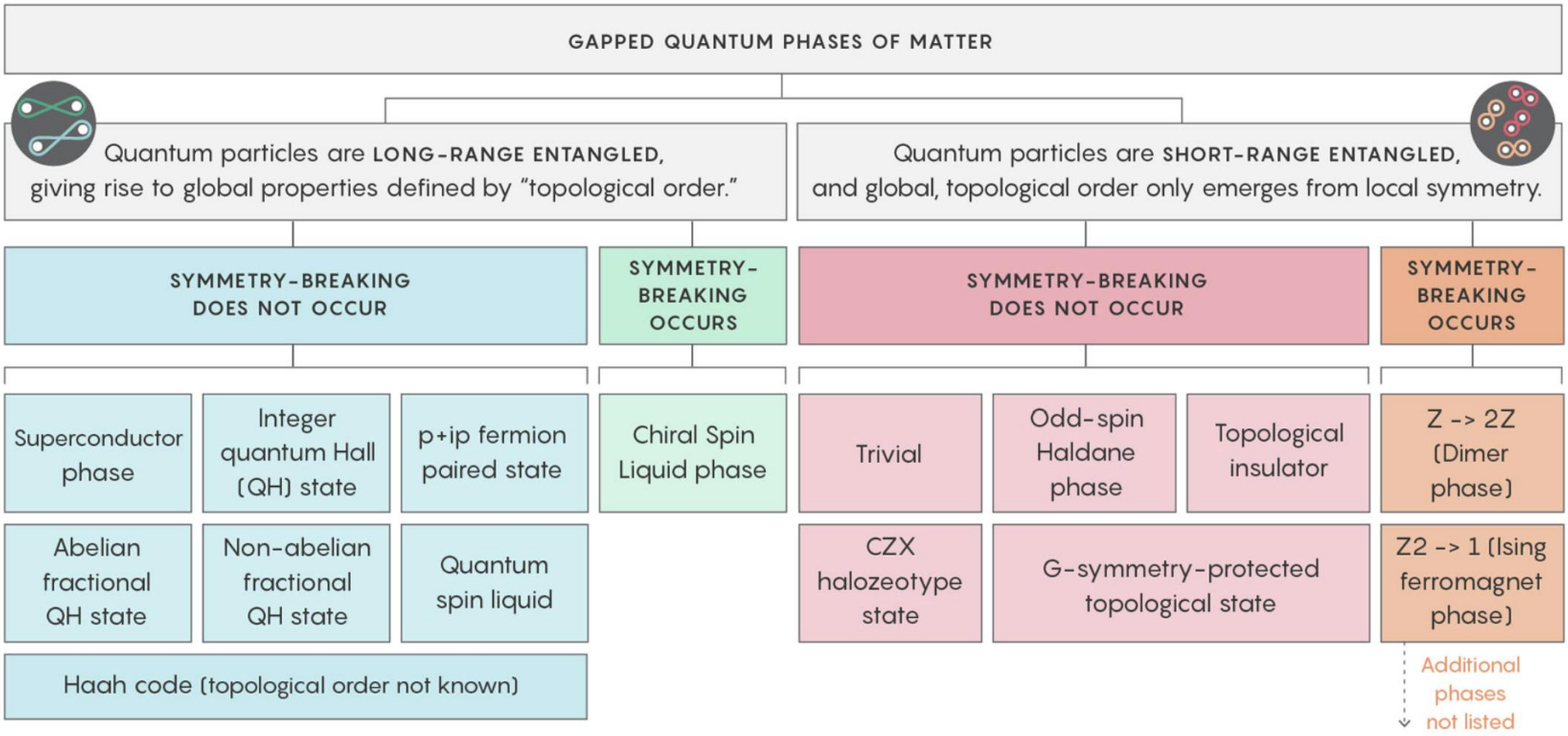
- Pierre Curie (1894):
  - “Phenomenon may exist in a medium having the same characteristic symmetry or the symmetry of a subgroup of its characteristic symmetry. What is necessary, is that certain elements of symmetry do not exist. Dissymmetry is what creates the phenomenon.”

- Landau paradigm:
  - Gapped phases are labeled by the (un)broken subgroups of the characteristic symmetry group of the Hamiltonian
  - Those phases can be detected by local order parameters which transform according to an irrep of the characteristic symmetry

- Generalized Landau paradigm (X.G. Wen, …):
  - Gapped phases are characterized by their entanglement structure



Phases of Matter



Lucy Reading-Ikkanda/Quanta Magazine, adapted from figure by Xiao-Gang Wen

The central conjecture

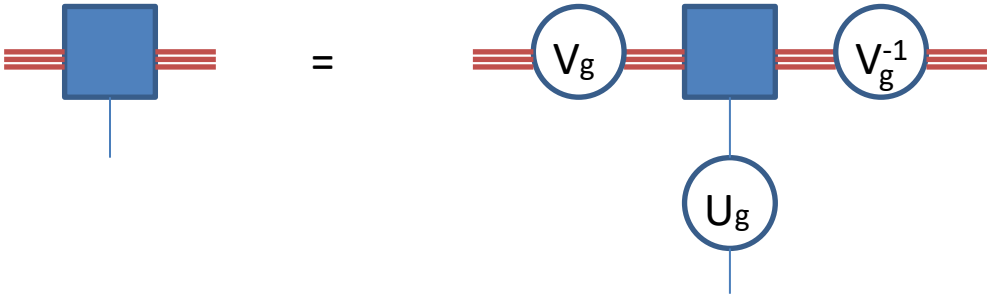
Tensor networks span the manifold of ground states. So a phase transition should happen exactly when the symmetry of the underlying tensors changes

phase transition ⇔ the tensor's symmetry changes

Different phases = different ways the symmetry acts on the entanglement legs

Symmetric tensors

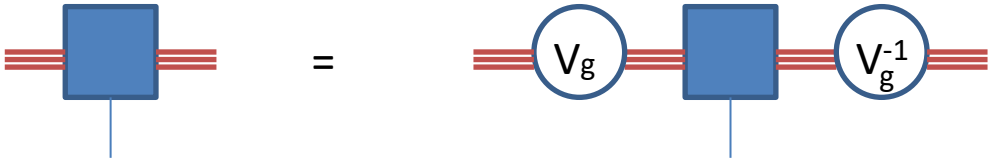
- Wigner-Eckert: Clebsh-Gordan coefficients yield a complete basis for symmetric tensors



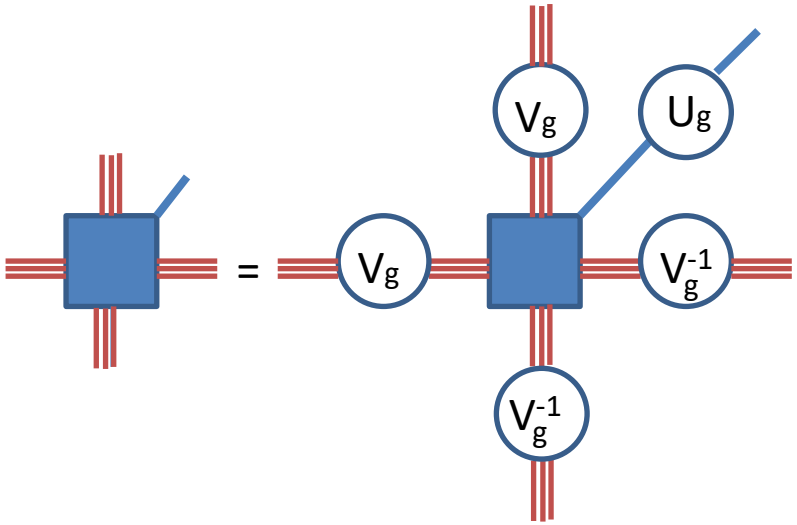
$$Rep_U \in Rep_V \otimes Rep_{\overline{V}}$$

ex: $1 \in \frac{1}{2} \otimes \frac{1}{2}$ -> V can be a projective representation!

- Purely “virtual” symmetry:
  - Reducible / non-injective tensor:

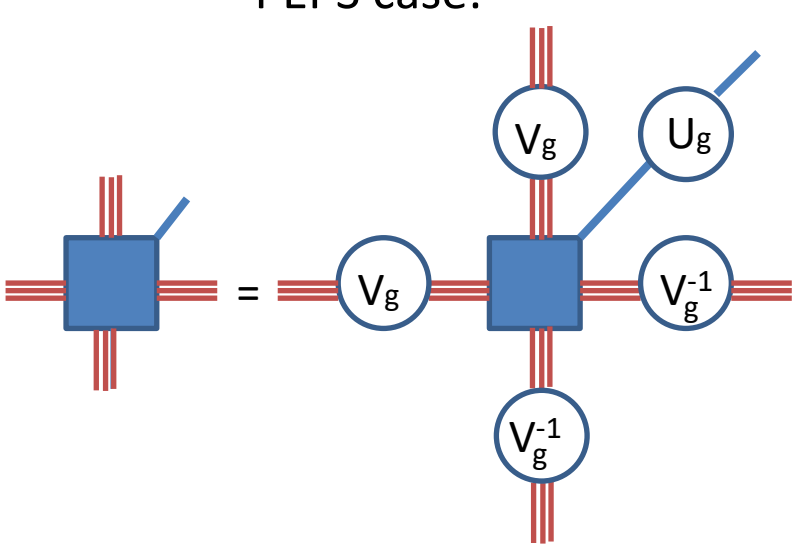


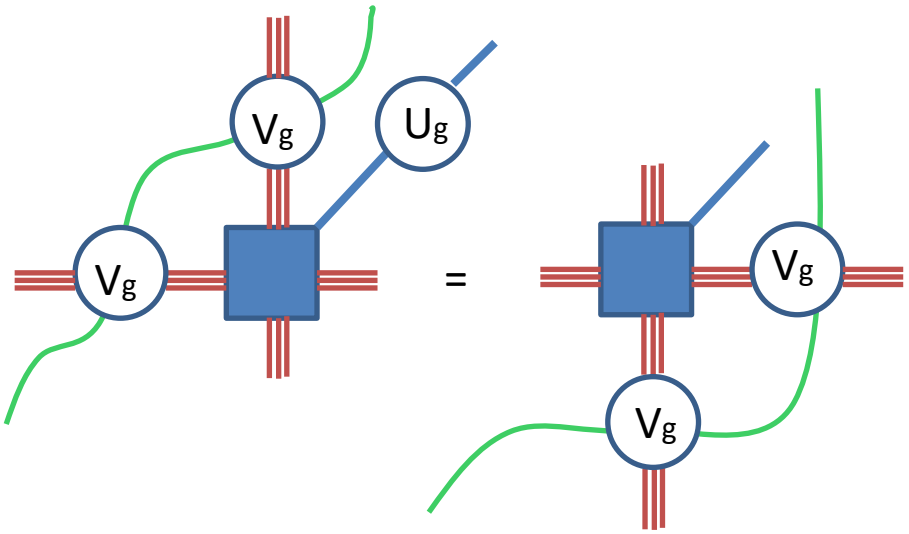
– PEPS case:



In words: a symmetry acting on the physical spins can be pushed onto the entanglement legs of the tensor. That is where the phase of matter is decided

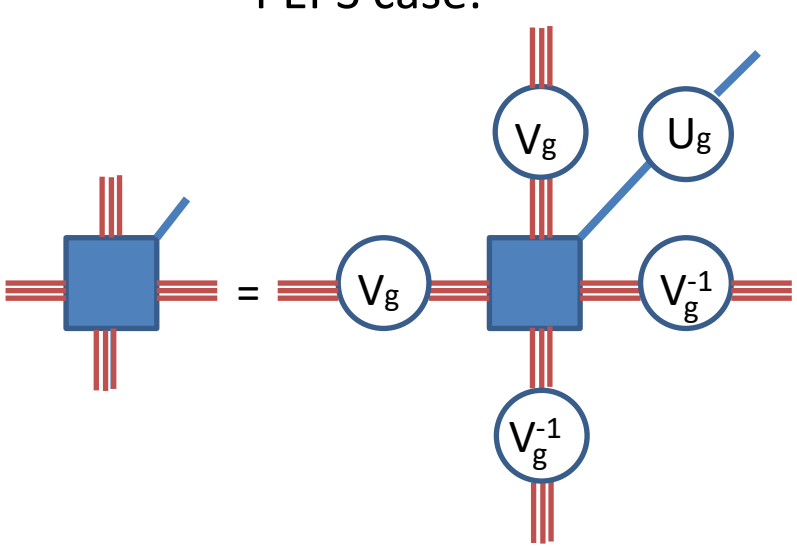
– PEPS case:

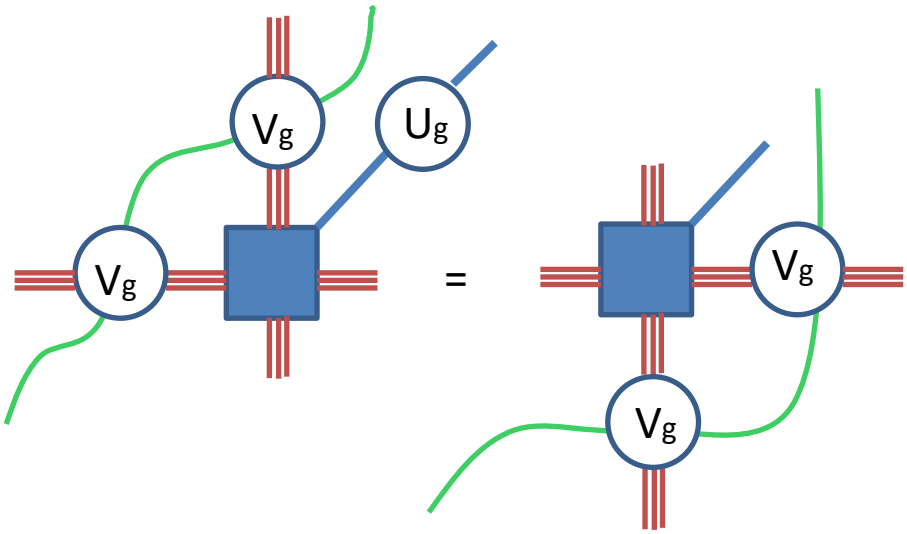




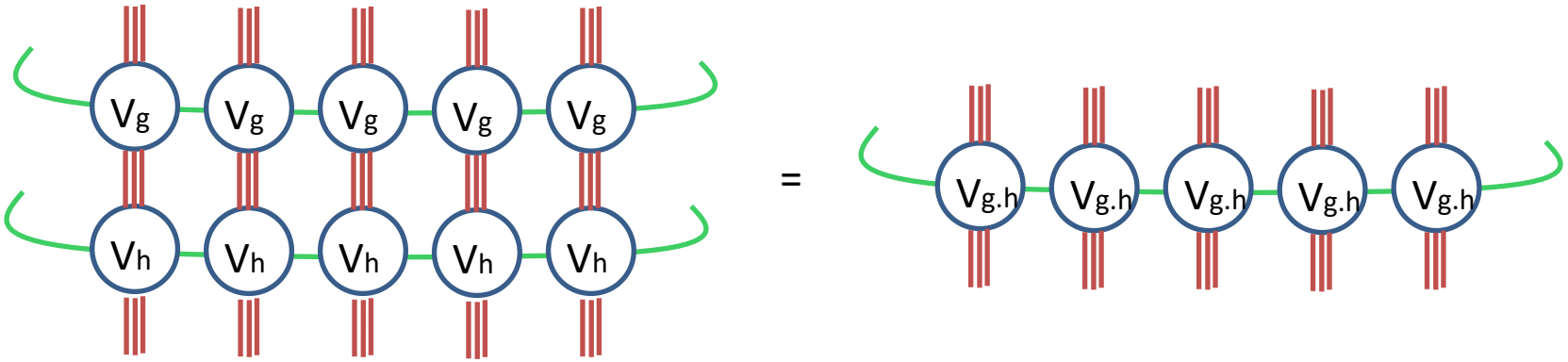
In words: a symmetry acting on the physical spins can be pushed onto the entanglement legs of the tensor. That is where the phase of matter is decided

– PEPS case:

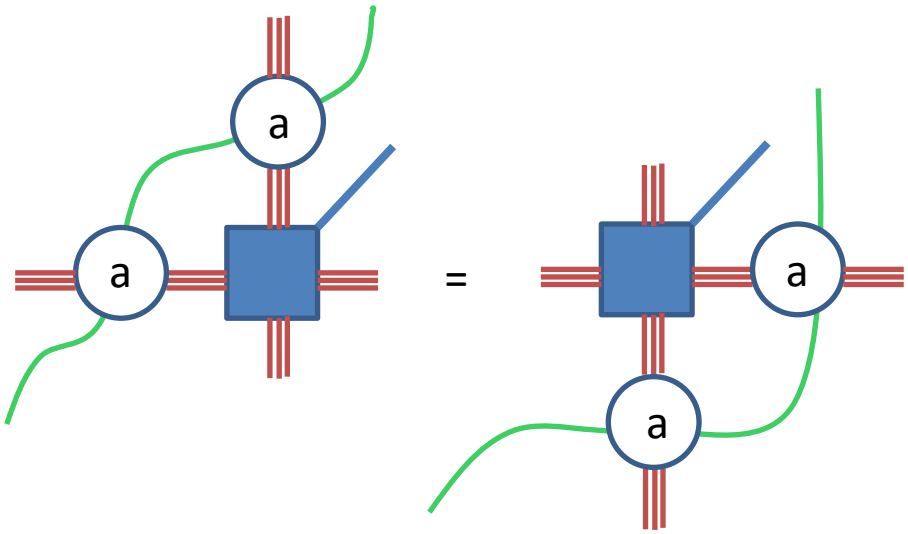




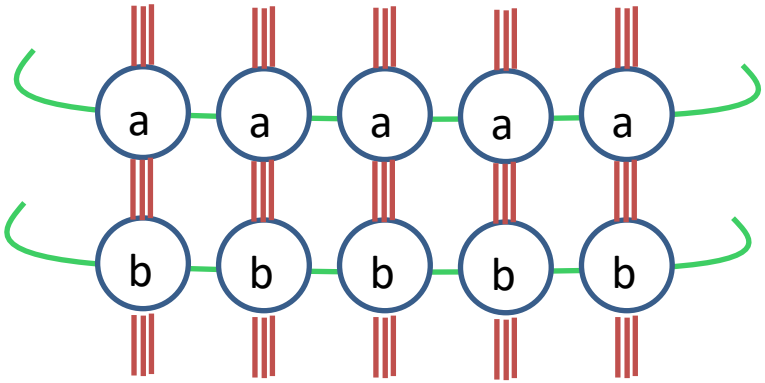
– MPO-symmetries: correlated, quantum group like action (“*defect*”)



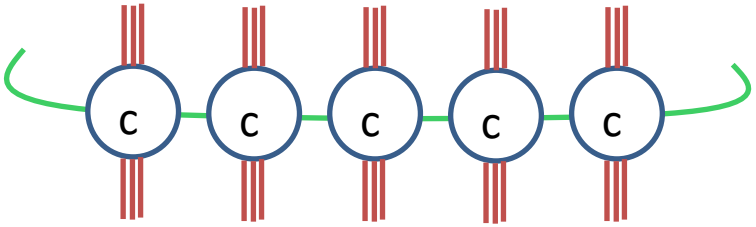
- Purely virtual symmetry: non-injective / MPO-injective case:



- Here the MPOs do not have to represent a group, but can represent a more general type of associative algbra (-> fusion categories)

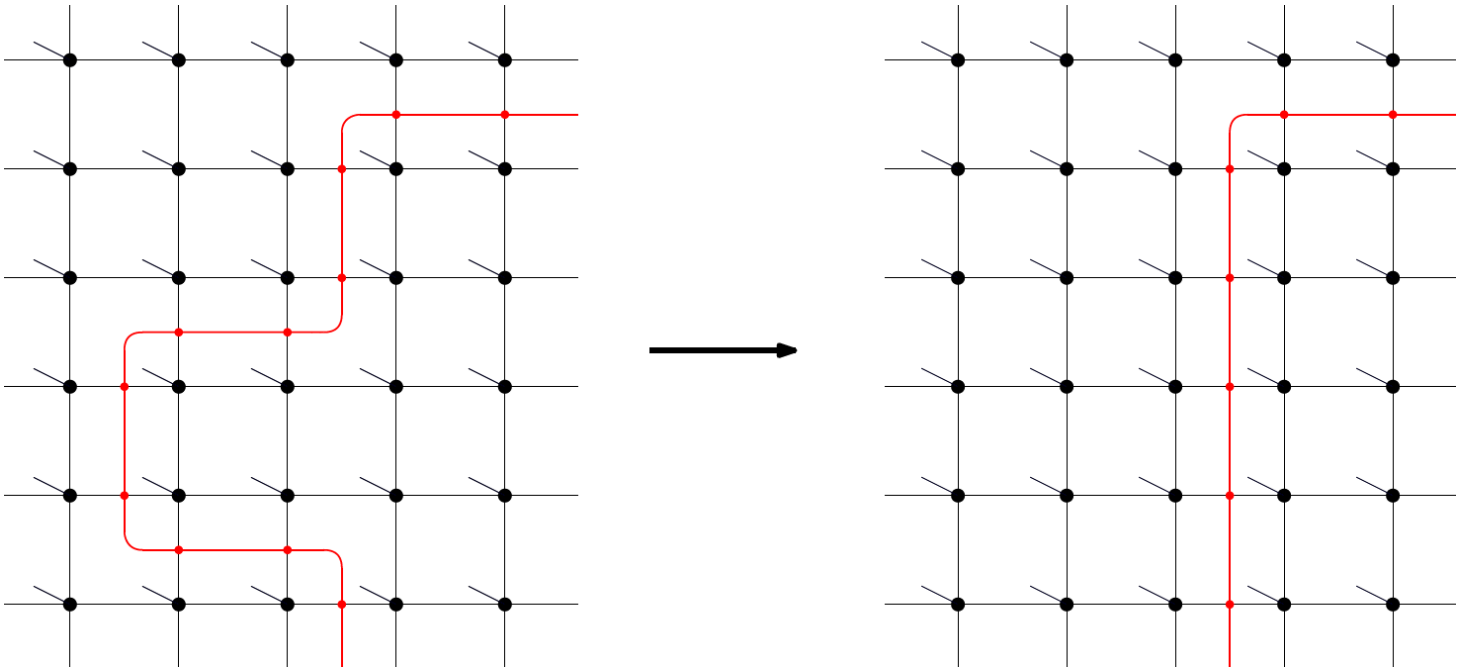


$$= \sum_c N_{ab}^c$$



Bultinck, Marien,Sahinoglu, Williamson, Haegeman, FV '15

- Topological order: a symmetry that acts only on the entanglement (virtual) legs of the tensors. The red string can be moved around freely and is invisible to any local measurement

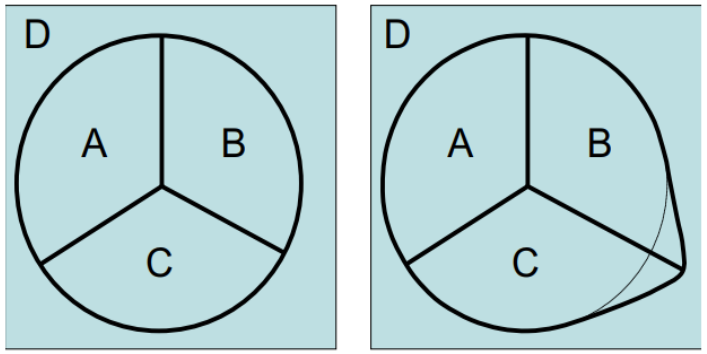


- This symmetry is what enables quantum error correction!
  - Toric code, string nets: exact tensor networks with this symmetry

Topological Entanglement Entropy

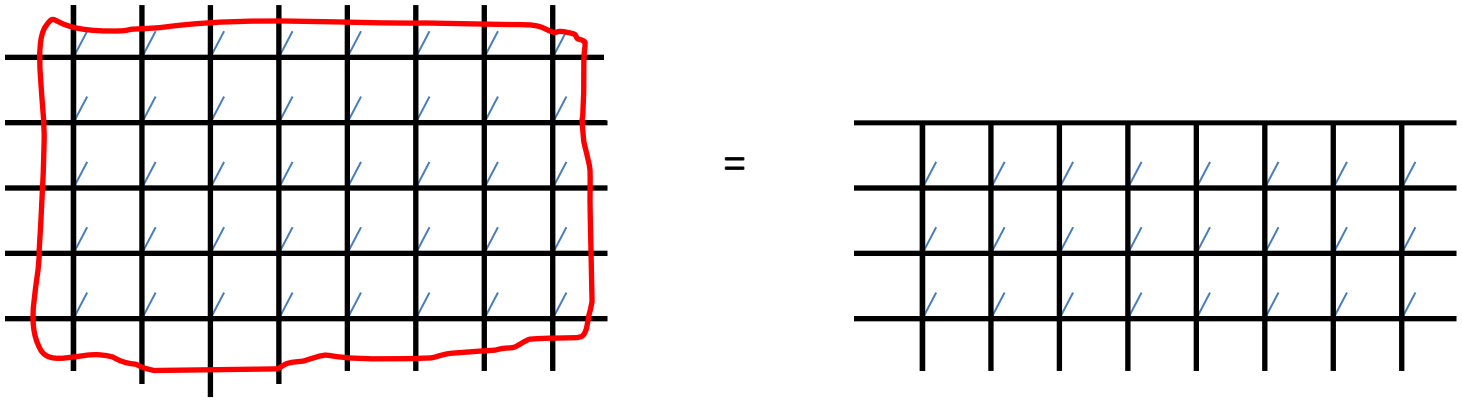
- *Kitaev, Preskill '06; Levin, Wen '06*: additive correction to area law for entanglement entropy in case of topological order

$$S(A) = c.\partial A - \log \sqrt{\sum_i d_i^2}$$



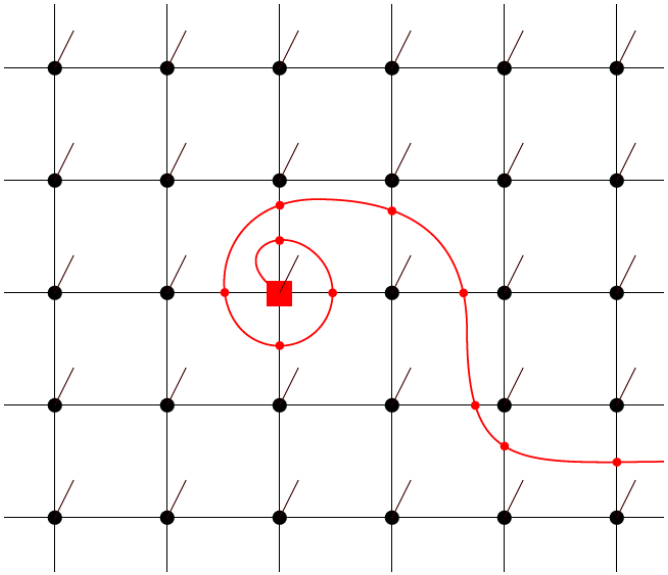
(a) (b)

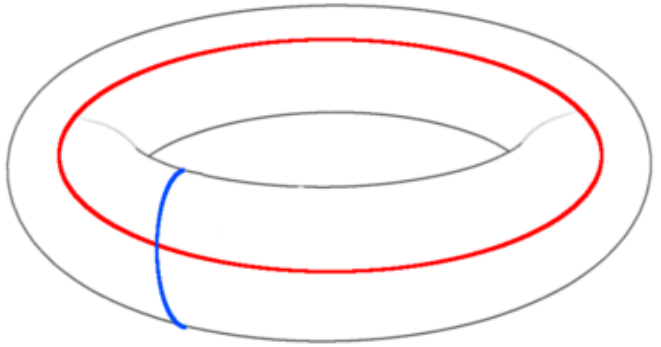
$$S_{\text{topo}} \equiv S_A + S_B + S_C - S_{AB} - S_{BC} - S_{AC} + S_{ABC}$$



- The Hilbert space of the *edge mode* is only supported on the symmetric MPO subspace: anyonic spin chains

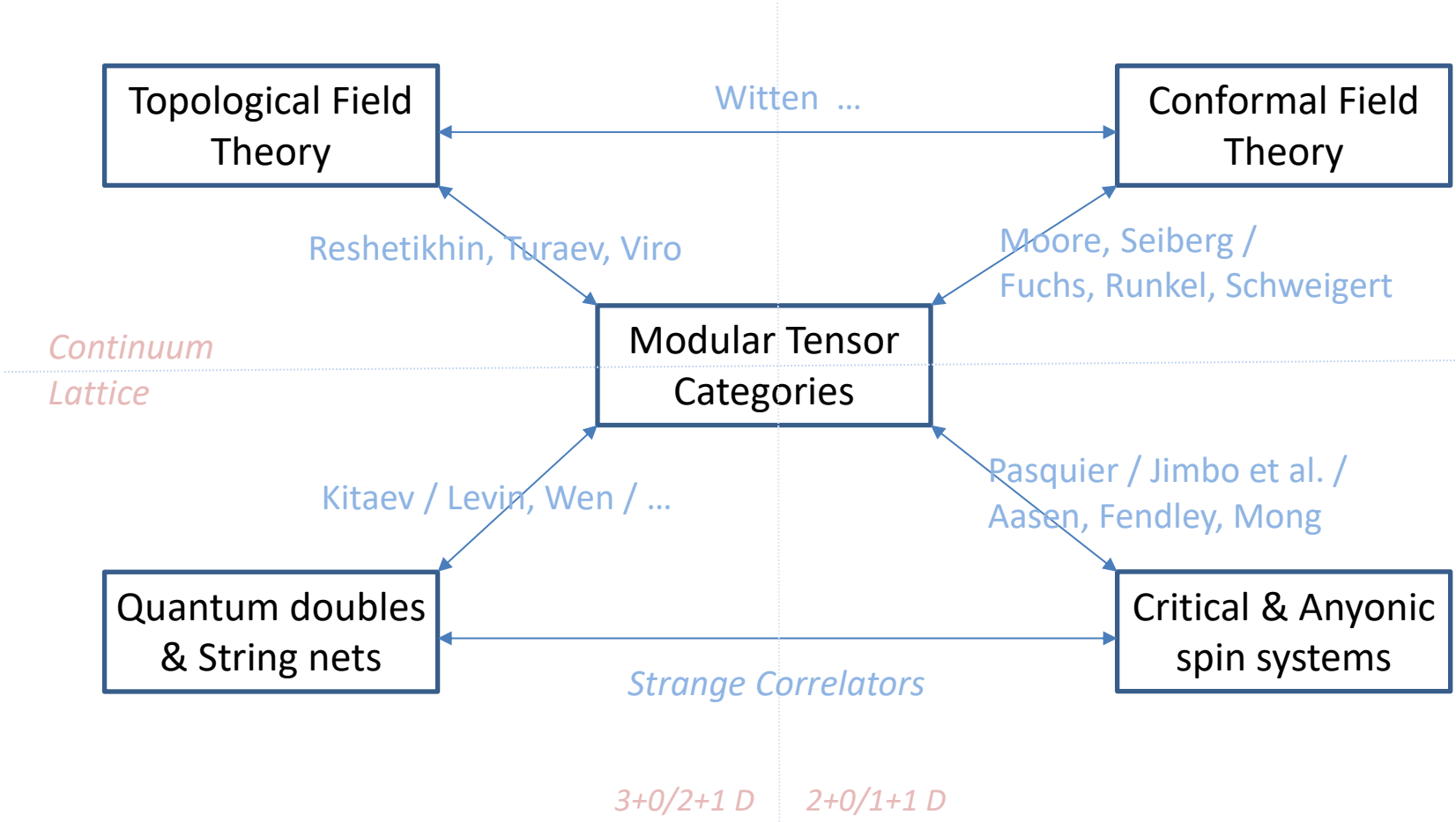
Topological sectors and anyons





- The Ocneanu ***tube algebra*** gives rise to idempotents (tensors) that describe:
  - Topological sectors in string nets (central idempotents)
  - Anyon excitations in string nets
  - Topological sectors in classical partition functions / topological boundary conditions in quantum spin chains (simple idempotents)

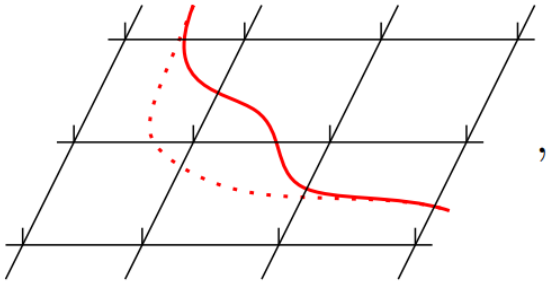
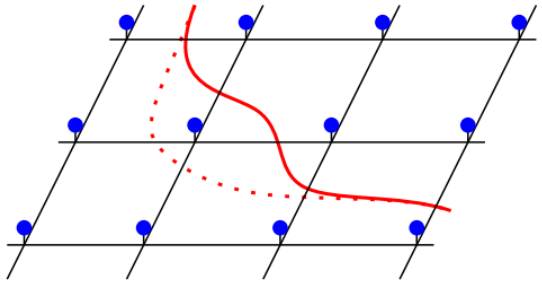
Categorical symmetries: TFT & CFT



In words: the same mathematics (modular tensor categories) describes topological phases, critical spin chains and conformal field theories

Duality defects / Strange correlators / SymTFT's

- From 2+1 to 2+0 by taking the overlap of the PEPS with a product state (or, more generally, with a PEPS in a different phase)

$|\psi_{\mathrm{SN}}\rangle =$  , $\langle\Omega|\psi_{\mathrm{SN}}\rangle =$ 

  - This construction yields a partition function of a classical stat. mech. model with non-invertible / categorical / MPO symmetries

- Equivalently: define a spin system with generalized symmetries as the edge modes of the bulk TFT (2+1 -> 1+1): anyonic spin chains

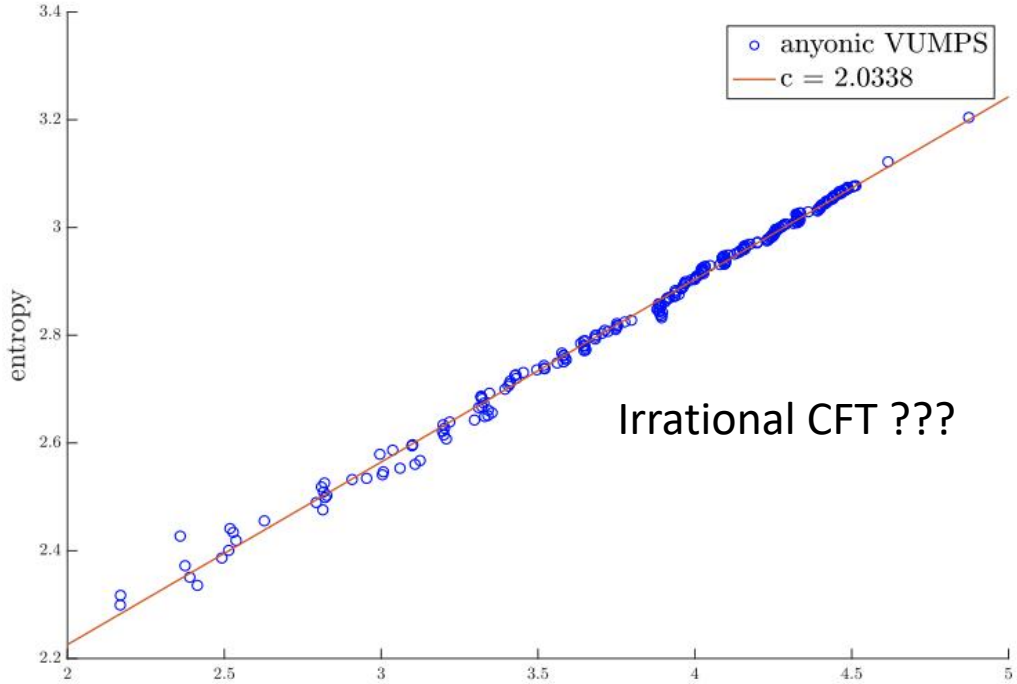
Vanhove, FV, et al. '18 ; …

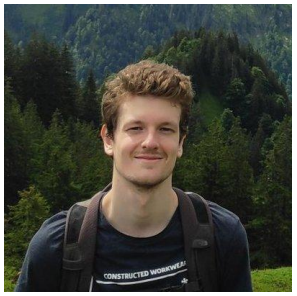
Critical Lattice Model for a Haagerup Conformal Field Theory

Robijn Vanhove,[1] Laurens Lootens,[1] Maarten Van Damme,[1] Ramona Wolf,[2,3]
Tobias J. Osborne,[3] Jutho Haegeman,[1] and Frank Verstraete[1]

Numerical Evidence for a Haagerup Conformal Field Theory

Tzu-Chen Huang,[1,*] Ying-Hsuan Lin,[2,*] Kantaro Ohmori,[3,*] Yuji Tachikawa,[4,*] and Masaki Tezuka[5,*]





nature physics

Article

https://doi.org/10.1038/s41567-025-02961-2

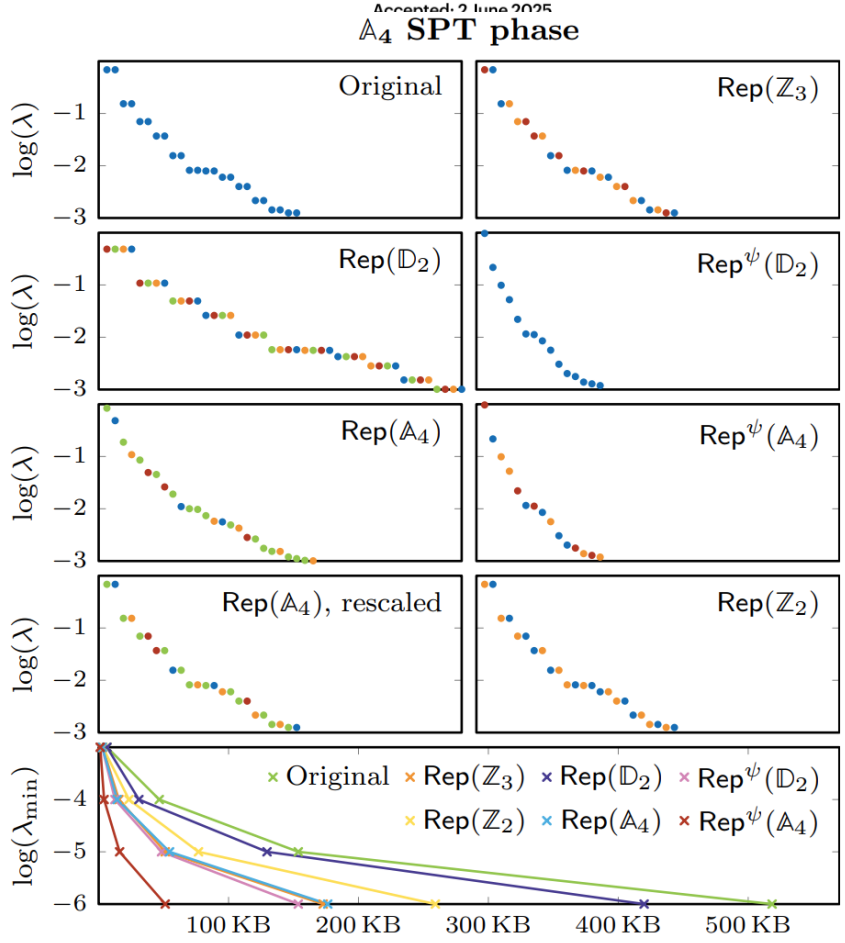
Entanglement and the density matrix renormalization group in the generalized Landau paradigm

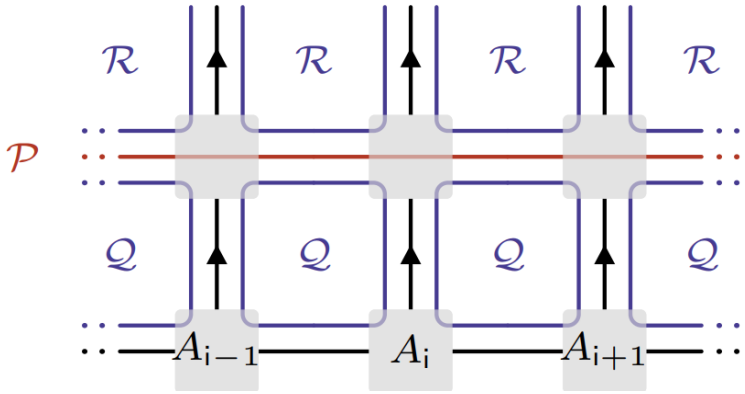
Received: 31 October 2024

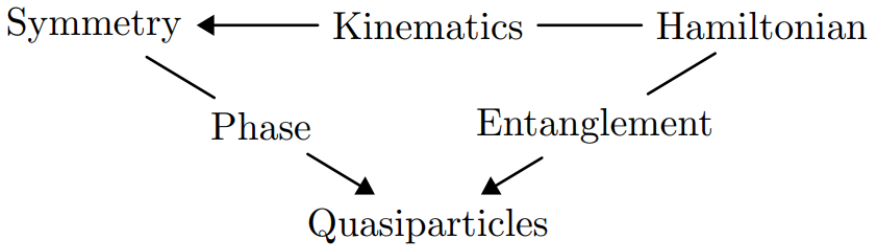
Accepted: 2 June 2025

Laurens Lootens [1,2] ✉, Clement Delcamp [3] & Frank Verstraete [1,2]









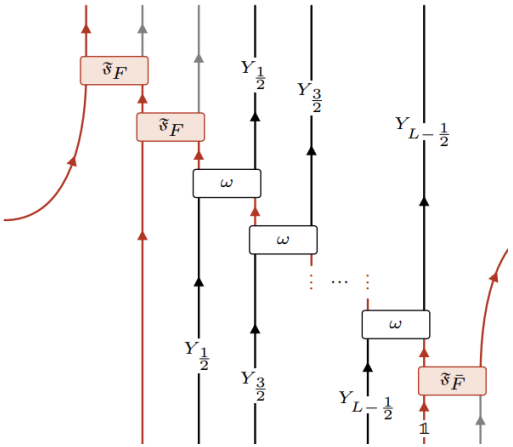
PHYSICAL REVIEW LETTERS **134,** 130403 (2025)

Editors' Suggestion

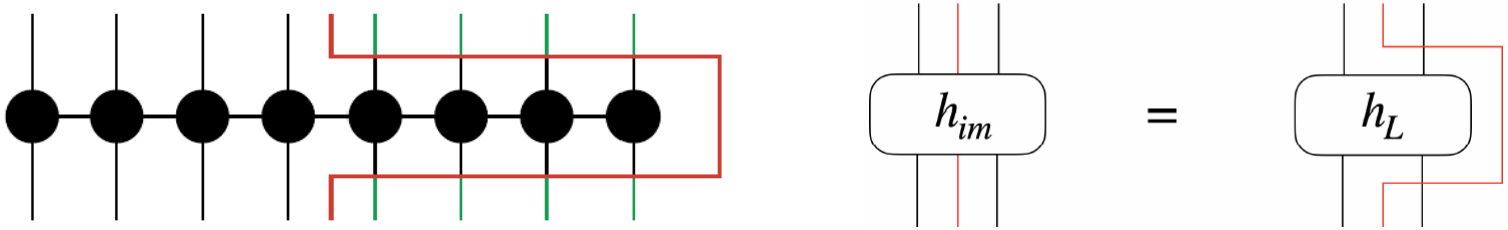
T S

Low-Depth Unitary Quantum Circuits for Dualities in One-Dimensional Quantum Lattice Models

Laurens Lootens,[1,2,*] Clement Delcamp,[2,3,†] Dominic Williamson,[4] and Frank Verstraete[1,2]



- Defects can be created by conjugating the Hamiltonian with this unitary; the Hilbert space dimension of a defect is equal to the virtual bond dimension!



nature physics

Article

https://doi.org/10.1038/s41567-026-03390-5

Perfect particle transmission through duality defects

Received: 22 November 2025

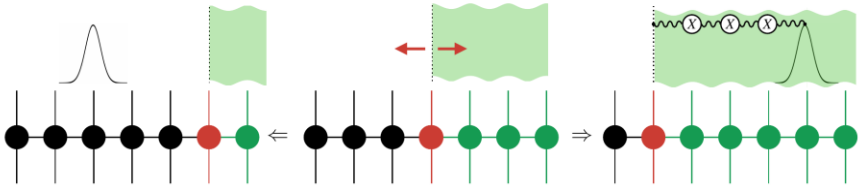
Accepted: 25 June 2026

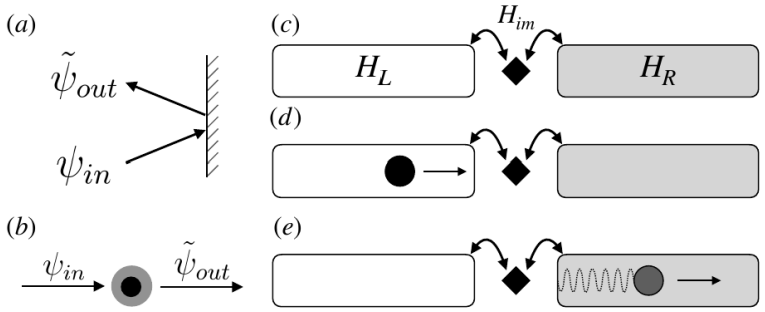
Atsushi Ueda[1], Vic Vander Linden[1], Boris De Vos[1], Laurens Lootens[2], Jutho Haegeman[1], Paul Fendley[3,4] & Frank Verstraete[1,2]

- An old puzzle: a fermion scattering off a magnetic monopole has nowhere to go

Callan, Rubakov '82; Polchinski '84; van Beest et al. '23

- Lattice version: a 'duality defect' that is itself a small tensor network (an MPO)
- A wavepacket is always perfectly transmitted, but comes out as a different, nonlocal particle: **fractionalization**





Take-home messages

1

Nature lives in a tiny corner of Hilbert space

Ground states of local Hamiltonians obey an area law: entanglement sits at the boundary

2

Tensor networks are built from entanglement

They parametrize exactly that corner, so an exponential problem becomes a polynomial one

3

A powerful computational tool

From counting problems and frustrated magnets to the Hubbard model at D = 28, the entropy of ice and simulating quantum computers

4

A new language for quantum matter

Phases, topological order and dualities are encoded in the symmetries of a single local tensor

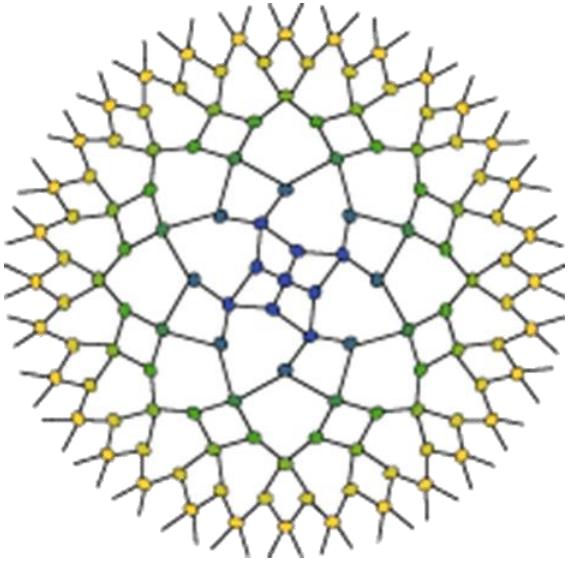
the unreasonable effectiveness of tensor networks in many-body physics?

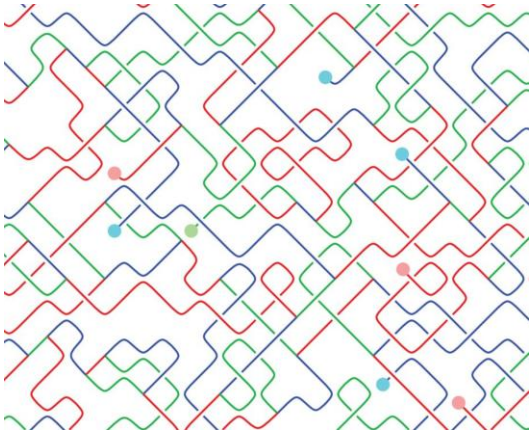
Real-Space Renormalization Group

Symmetries and dualities

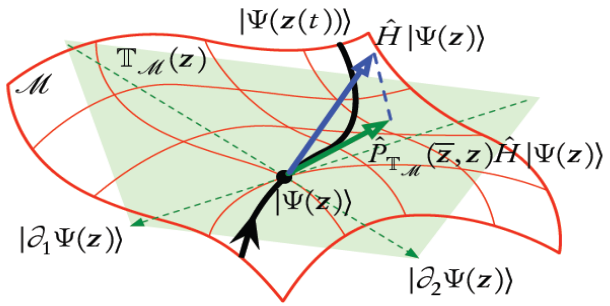
Tensor Networks

Quantum Field Theory
Statistical Mechanics





- High-Energy Physics
- Condensed Matter
- Cold Atoms



Fundamental theoretical ideas and computational aspects are not parallel lines of development, but are one and the same and must be fused with each other.

THE INTERNATIONAL BESTSELLER

WHY NOBODY UNDERSTANDS QUANTUM PHYSICS

The Story of the Science That Shapes Our World



FRANK VERSTRAETE & CÉLINE BROECKAERT